Notes on

Quantum Field Theory

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Contents

Chapter 1

Quantum field theory for children

These notes are based on an undergraduate course on quantum field theory (PHYS 187) taught by Professor Kevin Setter at Keck Science in spring 2026. The lectures were based on a book in progress by the instructor. Prof. Setter's philosophy is that there is a gap between the undergraduate curriculum and the presentation of quantum field theory to the graduate student. He develops the theory by "keeping things classical for as long as possible, then turning on quantum mechanics". He begins with a review of particle physics as a motivation, which I have kept in the appendix. The last sections are from things I learned during my final project.

Readings were assigned from Griffiths and Moore, and Peskin & Schroeder and Schwartz were supplements recommended by the course syllabus. Excerpts from Weinberg and deWitt were provided, as well as some of Prof. Setter's notes, based on texts like Hall. See the beginning of the appendix for more information on which books were used for respective portions of the appendix.

I called this chapter "quantum field theory for children" in part inspired the amusing title of Figueroa-O'Farrill's *Electromagnetic Duality for Children*. Prof. Setter's course gave me all the preparation that I needed to read and fully understand this text. Also, I was a child when I took the course (I was 17).

Things to be aware of: In many places, these notes deviate heavily from Prof. Setter's lectures; many of these excursions are outgrowths of several interesting conversations I've had with Prof. Setter outside of class. I have restricted many deviations to the chapter appendix. Also, some subsections are not written in my usual style. Particularly, the first few subsections of the section on spontaneous symmetry breaking are just toy models, which are nonetheless interesting, but not strictly necessary. Moreover, the notation is somewhat inconsistent across different sections, but everything should be consistent within each section.

1 Special relativity and Lorentz transformations of fields

1.1 Course-specific convention and index notation

- In this course, we use *natural units*, i.e. the system of units in which $\hbar = c = \varepsilon_0 = 1$.
- We use *Einstein summation notation*: if the same index appears exactly once in a superscript and exactly once in a subscript in any single term of an equation, we sum that term over the possible index values. For example, to denote a linear transformation R acting on a vector x, we write $x'^i = R^i_j x^j$.
- We work on Minkowski spacetime $\mathcal{M} = (\mathbb{R}^4, g)$, where g is the Lorentzian metric of signature $(+,-,-,-)$. In inertial coordinates,

  $$g_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1).$$

  The inverse metric $g^{\mu\nu}$ is defined by $g^{\mu\rho} g_{\rho\nu} = \delta^\mu_\nu$.

- $g_{\mu\nu}$ lowers indices, i.e. $v_\mu = g_{\mu\nu}v^\nu$. A quantity with a lowered index is called a *covector*, *one-form*, or *covariant vector*. Similarly, the contravariant metric raises indices: $v^\mu = g^{\mu\nu}v_\nu$.

1.2 Lorentz transformations and the active/passive interpretation

- Let V be a finite-dimensional vector space over a field K with basis $\mathfrak{B} = \{e_i\}$. This basis induces an isomorphism
$$K^n \xrightarrow{\sim} V, \qquad (v^1, \ldots, v^n) \mapsto v^i e_i.$$
  - An *active transformation* is a linear map $\tau \in \mathrm{End}(V)$ acting on vectors, $v \mapsto \tau v$, whose components in the basis $\mathfrak{B}$ are defined by $\tau e_i = \tau^j{}_i e_j$.
  - A *passive transformation* instead acts on the components of vectors. Given an invertible matrix $T = (T^i{}_j)$, the components transform as $v'^i = T^i{}_j v^j$. Requiring the vector itself to remain unchanged, $v = v^i e_i = v'^i e'_i$, determines the transformed basis $e'_i = (T^{-1})^j{}_i e_j$.
  - Intuitively, an active transformation moves the vector in a fixed basis, while a passive transformation changes the basis while leaving the vector unchanged.

- In the active interpretation, we consider mappings $\mathcal{M} \to \mathcal{M}$ described in a fixed coordinate system, where a point $x \mapsto x' = gx$. g is then a diffeomorphism of $\mathcal{M}$. In the passive interpretation, we consider g as a coordinate transformation so that x, x' denote the same point in two different coordinate systems.

- Define Lorentz distance (or spacetime interval) between two points $x, x' \in \mathcal{M}$ by $(\Delta s)^2 = g_{\mu\nu}(x^{'\mu} - x^\mu)(x^{'\nu} - x^\nu)$. The invariance group (*Poincare group*) consists of all transformations that leave this interval invariant, or alternatively, in the passive interpretation, which keep the formal expression for Lorentz distance unchanged by the transformation. This implies that x and x' are related by
$$x^{'\mu} = a^\mu + \Lambda^\mu_\nu x^\nu$$
where a^μ, Λ^μ_ν are constant and Λ satisfies $g_{\mu\nu}\Lambda^\mu_\alpha\Lambda^\nu_\beta = g_{\alpha\beta}$.

- Real 4×4 matrices satisfying this last condition, which may be recast in matrix form as $\Lambda^T g \Lambda = g$, are called *Lorentz transformations*.

- The set of Lorentz transformations forms the Lie group $O(1,3)$, called the *full Lorentz group*. The relation implies $\det \Lambda = \pm 1$ and $\Lambda^0_0 \in (-\infty, -1] \cup [1, \infty)$. The full Lorentz group consists of four disconnected pieces, depending on the sign of $\det \Lambda$ and whether Λ^0_0 is $\geq +1$ or ≤ -1. $\det \Lambda = -1$ means that Λ does not preserve orientation, i.e. that it corresponds to some reflection of the space. Such a transformation is said to be *improper*. $\Lambda^0_0 \leq -1$ means that the transformation flips time, and Λ is said to be *non-orthochronous*.

- When we refer to the *Lorentz group*, we specifically refer to the proper, orthochronous Lorentz group $SO^+(1,3)$.

The following example serves to show how index notation can be used.

Example. As an example of index notations, we show that the dot product of four-vectors is Lorentz-invariant: $u' \cdot v' = u \cdot v$ for any two 4-vectors u^μ, v^μ:
$$\begin{aligned}
u' \cdot v' &= u^{\mu'} v'_\mu \\
&= (\Lambda^\mu_\alpha u^\alpha)\left(g_{\mu\nu}\Lambda^\nu_\beta v^\beta\right) = u^\alpha v^\beta \left(\Lambda^\mu_\alpha \Lambda^\nu_\beta g_{\mu\nu}\right) \\
&= u^\alpha v^\beta \left(g_{\alpha\beta}\right) = u^\alpha v^\beta \left(g_{\alpha\beta}\right) \\
&= u \cdot v
\end{aligned}$$

- We define[1] a *tensor* to be a quantity with zero or more indices that transforms when one changes inertial reference frames in such a way that each upper index gets a factor of the Lorentz transformation Λ^μ_ν and each lower index gets a factor of the inverse Lorentz transformation matrix $(\Lambda^{-1})^\nu_\mu$. So, for example, a tensor with three indices would transform according to

$$T^{\alpha'}_{\beta\mu} = \Lambda^\alpha_\nu (\Lambda^{-1})^\sigma_\beta (\Lambda^{-1})^\eta_\mu T^\nu_{\sigma\eta}$$

  A tensor with n indices is an n-th rank tensor, and such a tensor has 4^n components (in 4D spacetime). We can be more specific and speak of tensors of type (p, q) (that is, tensors with p raised indices and q lowered indices), in which case the above tensor is of type $(1, 2)$.

- A spacetime *scalar field* $\phi(x)$ is defined by the property that its value is invariant under Lorentz transformations. In the active view, the field configuration is shifted as $\phi(x) \to \phi'(x) = \phi(\Lambda^{-1}x)$. In the passive view, the scalar field transforms trivially as $\phi'(x') = \phi(x)$.

- Concretely, fields transform according to their tensor character, so a four-vector field $A^\mu(x)$ would transform as

$$A^{\mu'}(x') = \Lambda^\mu_\nu A^\nu(x)$$

  while some rank 2 tensor field $T'^{\mu\nu}(x)$ would transform as

$$T^{\mu\nu'}(x') = \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}(x)$$

- The partial derivative operator $\partial_\mu := \frac{\partial}{\partial x^\mu}$ is a local section of the tangent bundle. To raise its index, we have $\partial^\mu = g^{\mu\nu}\partial_\nu$. When transforming from a 3D scalar field to a 3-vector field, e.g. the relationship between electric field of voltage, this corresponds to

$$E^i(\vec{x}) = \partial^i V(\vec{x}) = -\partial_i V(\vec{x}) = -\frac{\partial}{\partial x^i} V(\vec{x})$$

1.3 Representations of Lorentz transformations

- More generally, if we write transformations of fields as $\Phi \to M(\Lambda)\Phi$, we only require that M be an n-dimensional *representation* of the Lorentz group.

- It is generally true that one can find matrix representations of a continuous group by finding matrix representations of the generators of the group satisfying the proper commutation relations. For the *Lorentz algebra*, the commutation relations are

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho})$$

  Any matrices that are to represent this algebra must obey these commutation rules.

- Warning: In physics, the generators $J^{\mu\nu}$ are taken to be Hermitian operators, so their commutation relations include explicit factors of i. Strictly speaking, the Lie algebra $\mathfrak{so}(1,3)$ is defined using real (or equivalently anti-Hermitian) generators with no such factors. Thus, the operators uused here furnish a representation of the *complexified Lie algebra*[2] $\mathfrak{so}(1,3)_\mathbb{C}$, rather than a literal real form of $\mathfrak{so}(1,3)$.

- An example of such a representation, used in Peskin & Schroeder, is the 4×4 matrices

$$(\mathcal{J}^{\mu\nu})_{\alpha\beta} = i(\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha)$$

  Here, μ and ν label which of the six matrices you want, while α and β label the components. These are the matrices that act on ordinary Lorentz 4-vectors.

- This course relabels the generators as follows; the *boost generators* are $K^i = J^{0i}$, and the *rotation generators* are $J^i = \frac{1}{2}\epsilon^{ijk}J^{jk}$. Note that, explicitly, $J^1 = J^{23}$, $J^2 = J^{31}$, $J^3 = J^{12}$; the $\frac{1}{2}$ compensates for the double counting in the sum.

[1]Warning to the mathematician: the following is a working definition of a tensor for physics in flat spacetime. Do not faint!
[2]See appendix.

2 Relativistic classical field theory

2.1 Euler-Lagrange equations

- In local field theory, the Lagrangian can be written as the spatial integral of a *Lagrangian density* $\mathcal{L}$ which is a function of one or more fields $\phi(x)$ and their derivatives $\partial_\mu \phi$. Thus we have the *action*

$$S = \int \mathcal{L}(\phi, \partial_\mu \phi) d^4 x$$

  Action is dimensionless in natural units. In field theory, the term "Lagrangian" is used to refer to the Lagrangian density.

- The corresponding *Euler-Lagrange equation of motion for a field* is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0$$

- Let us decompose the Lagrangian density into a quadratic (free) part and higher-order interaction terms, $\mathcal{L} = \mathcal{L}^{(2)} + \mathcal{L}^{(\geq 3)}$. A field is *dynamical* (or propagating) if its quadratic Lagrangian contains time derivatives. The derivative-containing quadratic terms define the *kinetic term.*

- We canonically normalize our fields such that the kinetic term has its standard coefficient; for example, for a real scalar field, the canonically normalized quadratic Lagrangian is $\mathcal{L}^{(2)} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2$ (i.e. the standard coefficient is $\frac{1}{2}$). Quadratic terms without derivatives, such as $\frac{1}{2} m^2 \phi^2$, are mass terms; although bilinear, they are not kinetic terms. Together with the derivative term, they determine the free equation of motion and inverse propagator. All terms cubic or higher in the fields are called *interaction terms*; these govern departures from free propagation.

- In general, a (classical) *field theory* is a particular collection of classical fields free to fluctuate throughout spacetime in various configuations and an action governing the fluctuations of the field. A *scalar field theory* is a field theory of scalar fields.

2.2 Scalar fields in 4D, Klein-Gordon equation

- An important case is that of the *real scalar field in 4d*: if $\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$, the equations of motion are the *Klein-Gordon equation*

$$(\Box + m^2)\phi = 0$$

  where we use the *D'Alembertian* $\Box = \partial^\mu \partial_\mu$.

- For a *complex scalar field in 4d*, $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$. ϕ and ϕ^*, treated as independent fields when varying, each satisfy the Klein-Gordon equation.

2.3 Noether's theorem

- A *symmetry* of a classical field theory is a transformation of fields (and possibly coordinates) $x^\mu \mapsto x'^\mu(x)$, $\phi_i(x) \mapsto \phi_i'(x')$ such that $S[\phi'] = S[\phi]$ for all field configurations satisfying appropriate boundary conditions. Equivalently, a transformation is a symmetry iff it maps solutions of the Euler-Lagrange equations to other solutions.

- The action is invariant[3] if the Lagrangian density transforms so that $\delta_\epsilon \mathcal{L} = \partial_\mu K^\mu$ for some functional K^μ, with the boundary term vanishing under the assumed boundary conditions. Special cases include:

[3]Sometimes, one might (loosely) refer to an "invariant Lagrangian"; this does not mean that the Lagrangian itself is invariant, but is rather shorthand for "Lagrangian density functional such that, when integrated over spactime with appropriate boundary conditions, produces an invariant action."

  - Strict invariance of the Lagrangian, $\mathcal{L}' = \mathcal{L}$
  - The Lagrangian density transforms as a scalar field
  - The Lagrangian density shifts by a 4-divergence ($\mathcal{L}'[\Phi, \partial_\mu \Phi] = \mathcal{L}[\phi, \partial_\mu \Phi] + \partial_\mu K^\mu$ with K^μ vanishing at the boundary of integration region)

- A *continuous symmetry* is a smooth one-parameter group of diffeomorphisms ϕ_ϵ on the space of field configurations (indexed by ϵ) such that ϕ_0 is the trivial symmetry, $\phi_{\epsilon_1} \circ \phi_{\epsilon_2} = \phi_{\epsilon_1+\epsilon_2}$, and the map $(\epsilon, x) \mapsto \phi_\epsilon(x)$ is smooth. For global symmetries, the parameter ϵ is a constant in spacetime. The infinitesimal variation is defined by $\delta_\epsilon \phi_i = \epsilon \Delta\phi_i$.

- *Noether's theorem* states the following. Let the action be invariant under a continuous global symmetry, meaning $\delta_\epsilon \mathcal{L} = \partial_\mu K^\mu$. Then the *current*

$$j^\mu = \sum_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \Delta\phi_i - K^\mu$$

  satisfies the identity

$$\partial_\mu j^\mu = \sum_i \frac{\delta S}{\delta \phi_i} \Delta\phi_i$$

  Therefore, on-shell (i.e. for solutions of the Euler-Lagrange equations), $\partial_\mu j^\mu = 0$, implying the conserved *charge* $Q = \int d^3x j^0$ is time-independent.

2.4 Discrete symmetries

- The (unitary) *parity transformation* inverts all spatial coordinates (by reflection through the origin), i.e. it is the transformation $\vec{x} \mapsto -\vec{x}$. All fundamental interactions of elementary particles, with the exception of the weak interaction, are symmetric under parity transformation.

- *Time reversal* is what it sounds like ($t \mapsto -t$).

- For a collection of fields $\Phi(x)$ on spacetime, a *charge conjugation transformation* $C : \Phi \to \Phi^C$ is a discrete, involutive ($C^2 = 1$) map with $\mathcal{L}(\Phi^C, \partial\Phi^C) = \mathcal{L}(\Phi, \partial\Phi)$ such that for every conserved Noether charge Q associated with an internal $U(1)$-type symmetry, $C : Q \mapsto -Q$. Thus C acts as an automorphism of the classical phase space preserving the symplectic structure while reversing specific internal quantum numbers.

- Concretely, suppose the theory possesses a complex field ϕ transforming under a global $U(1)$ ($\phi \to e^{i\alpha}\phi$). Then charge conjugation acts as complex conjugation, and the corresponding Noether current $j^\mu = i(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*)$ transforms as $j^\mu \overset{C}{\to} -j^\mu$ hence reversing the associated conserved charge; C exchanges a representation with its complex-conjugate representation under the internal symmetry group.

- Note that real fields are trivially C-invariant. A complex scalar field may be regarded as two real fields, and vice versa. The latter is important whenever a field transforms in a complex representation of an internal symmetry (e.g., typically, $U(1)$, $SU(2)$, $SU(3)$). For example, two real scalar fields $\phi_1(x), \phi_2(x)$ with Lagrangian $\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2$ has internal symmetry group $O(2)$, acting by rotations in the two-dimensional real field space; if we define $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$, the $SO(2)$ rotation becomes a $U(1)$ phase transformation. Thus the complex description is a convenient representation of the two-component real multiplet.

- The *CPT theorem* says that any Lorentz invariant local quantum field theory with a Hermitian Hamiltonian must have have CPT symmetry, i.e. that the simultaneous transformations of charge conjugation, parity transformation, and time reversal constitute a fundamental symmetry of physical laws.

2.5 Energy-momentum tensor

- For most field theories we care about (like that of a real scalar field), the fields in every little region of possess 4-momentum. There are four separate local conservation laws, so four separate conserved 4-currents. Altogether the $4 \times 4 = 16$ components of these 4-vectors are packaged together into a single object, the (canonical[4]) *energy-momentum tensor* $T^{\mu\nu}$:

$$j^{\mu}_{(\text{energy})} = T^{\mu 0} = T^{0\mu}, \quad j^{\mu}_{(x\text{-momentum})} = T^{\mu 1} = T^{1\mu}, \quad \text{and so on.}$$

- The formula for $T^{\mu\nu}$ given the Lagrangian density $\mathcal{L}$ is

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)}(\partial^\nu \phi^i) - g^{\mu\nu}\mathcal{L}$$

  where we sum over all fields ϕ^i; of course, this satisfies $\partial_\mu T^{\mu\nu} = 0$ on shell.

[4]The physically prefered energy-momentum tensor is defined by variation of the action with respect to the spacetime metric. One can make the canonical energy-momentum tensor symmetric by adding an identically conserved improvement term, which results in the Belinfante-Rosenfeld tensor, which agrees (up to improvement terms) with the general energy-momentum tensor in flat spacetime. For our purposes, however, the definition given is sufficient.

3 Spinor fields

This chapter will provide a working construction of the topics. We keep things mathematical and do physics in the next section. There is much deeper underling mathematics, far beyond that required of a traditional physics course. A further treatment has been attempted in the appendix.

3.1 Motivation and background

The motivation for this section is as follows. While a general procedure for finding finite-dimensional representations of the Lorentz group is not difficult, we are mainly interested in those corresponding to spin 1/2, as these are required to describe fermions. So far, the fields we have constructed are insufficient. So far, a particle at rest is an excitation of a normal mode of a scalar field with $|\vec{k}| = 0$ (with angular momentum $L_z = 0$, since $\delta_\epsilon \phi = \epsilon \left(-\frac{\partial \phi}{\partial \varphi}\right)$ under an infinitesimal rotation). However, obviously, fermions have spin, so they cannot be described using a scalar field; in fact, the only spinless particle of the standard model is the Higgs boson. One might guess that a fermion field is a 4-vector field, but quantized excitations of normal modes of 4-vector fields are *spin-1* particles.

We will need to keep in mind the *Pauli matrices*:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

More compactly, $[\sigma_j, \sigma_k] = 2i\epsilon_{jkm}\sigma_m$. These commutation relations make the Pauli matrices the generators of a representation of the Lie algebra $(\mathbb{R}^3, \times) \cong \mathfrak{su}(2) \cong \mathfrak{so}(3)$. They also satisfy the anticommutation relations $\{\sigma_j, \sigma_k\} = 2\delta_{jk}I$, making them the generators of a representation of the Clifford algebra for $\mathbb{R}^3$, denoted $Cl(\mathbb{R})$.

3.2 Representations of Lorentz transformations for spin 1/2

- The motivation is as follows. To build a relativistic equation for a field that is first-order in derivatives, Dirac's idea was to factor the Klein-Gordon operator $\Box + m^2\psi$ into $(i\gamma^\mu\partial_\mu - m)(i\gamma^\mu\partial_\nu + m)\psi$. The condition for this to be true is that the γ^μ are elements of the following.

- Define the *Dirac algebra* to be the Clifford algebra $Cl_{1,3}(\mathbb{C})$. The *gamma matrices* are a set of conventional matrices $\{\gamma^0, \gamma^1, \gamma^2, \gamma^3\}$, satisfying the anticommutation relations

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$$

  These generate a matrix representation of the Dirac algebra.

- The *Feynman slash notation* simplifies things as follows: if A is a covariant vector, we define $\not{A} = \gamma^\mu A_\mu$.

- This course uses the *Weyl* or *chiral representation*:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

- Infinitesimally, an Lorentz transformation can be written $\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu$, where $\omega^\mu_\nu \in \mathfrak{so}(1,3)$ is an infinitesimal parameter satisfying $\omega_{\mu\nu} = -\omega_{\nu\mu}$. This acts on a field as $\psi \mapsto \psi - \frac{1}{2}\omega_{\mu\nu}S^{\mu\nu}\psi$, where the matrices $S^{\mu\nu}$ are generators of Lorentz transformations on the field space, satisfying the Lorentz algebra commutation relations from the previous section.

- Lorentz covariance of $(i\not{\partial} - m)\psi = 0$ forces

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]I$$

- In the Weyl representation, the boost and rotation generators are

$$S^{0i} = \frac{i}{4}[\gamma^0, \gamma^i] = -\frac{1}{2}\begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}$$

  and

$$S^{ij} = \frac{i}{4}[\gamma^i, \gamma^j] = \frac{1}{2}\epsilon^{ijk}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix} \equiv \frac{1}{2}\epsilon^{ijk}\Sigma^k$$

  Note that the rotation generator S^{ij} is just the three-dimensional spinor transformation matrix $\frac{1}{2}\epsilon^{ijk}\sigma^k$ replicated twice.

- Recall that the rotation algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ has irreducible representations labeled by a single number j, with dimension $2j+1$.[5] In particular, the spin 1/2 representation is 2d with generators $J^i = \frac{1}{2}\sigma^i$.

3.3 Spinors and their transformations

- Note that $SO^+(1,3)$ is not simply connected.[6] As such, there exist projective representations of $SO(1,3)$ in which a 2π rotation acts as -1 rather than identity.[7] To treat them as honest linear representations, we pass to the universal cover $SL(2,\mathbb{C}) \to SO^+(1,3)$, which is simply connected.

- *Spinor representations* are the representations of $SL(2,\mathbb{C})$ in which the nontrivial element of the kernel acts noontrivially.

- Physically, define $A^i = \frac{1}{2}(J^i + iK^i)$, $B^i = \frac{1}{2}(J^i - K^i)$, which then satisfy $[A^i, A_j] = i\epsilon^{ijk}A^k$, $[B^i, B^j] = i\epsilon^{ijk}B^k$, $[A^i, B^j] = 0$. Complexifying the Lorentz group,[8] this gives us $\mathfrak{so}(1,3)_{\mathbb{C}} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$.

- *Dirac spinors* are elements of a 4-dimensional complex vector space $\left(\frac{1}{2}, 0\right) \oplus \left(0, \frac{1}{2}\right)$ representation of the Lorentz group.

- The Dirac spinor furnishes a reducible repreesentation of the group of Lorentz transformations; in the Weyl representation, a Dirac spinor ψ has four complex components, the top two of which consist of a *left-handed Weyl spinor* ψ_L and the bottom two components a *right-handed Weyl spinor* ψ_R:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

- We may alternatively define a Dirac spinor as a four-component field ψ that transforms under boosts and rotations according to the above generators.

- The boost generators S^{0i} are not Hermitian, and thus our implementation of boosts is not unitary; the Lorentz group, being noncompact, has no faithful, finite-dimensional representations that are unitary. But that does not matter to us, since ψ is a classical field, not a wavefunction.

- To write a Lagrangian for the Dirac theory (these notions will be clarified in the next section), we must figure out how to multiply two Dirac spinors to form a Lorentz scalar. Define[9] $\overline{\psi} = \psi^\dagger\gamma^0$.

[5] Finite-dimensional irreducibility implies the existence of a highest-weight state $|j\rangle$ annihilated by the raising operator J_+. Repeated action of the lowering operator J_- generates a finite ladder of states with J_3 eigenvalues $m = j, j-1, \cdots, -j$. Closure of the algebra forces j to be integer or half-integer and fixes dimension to be $2j+1$.

[6] $SO^+(1,3)$ and $SO(3)$ are of the same homotopy type, so $\pi_1(SO^+(1,3)) \cong \pi_1(SO(3))$. Topologically, $SO(3) \cong S^3/\{\pm 1\} \cong \mathbb{Z}_2$. The second to last isomorphism comes from the fact that, for $SO(3)$, rotating by θ about an axis n is the same as rotating by $2\pi - \theta$ around $-n$.

[7] An aside, jumping forward–projective representations of $SO(3)$ are called *fermionic*, and faithful representations are called *bosonic*, for reasons which will be apparent.

[8] Recall that given a real Lie algebra, there is a one-to-one correspondence between complex representations of the algebra and the representations of its complexification.

[9] Note that we need to use this, rather than the obvious guess, $\psi^\dagger\psi$, because, under a Lorentz boost, the latter becomes $\psi^\dagger\Lambda^\dagger_{\frac{1}{2}}\Lambda^{-1}_{\frac{1}{2}}\psi$; unitarity would give $\Lambda^\dagger_{\frac{1}{2}} = \Lambda^{-1}_{\frac{1}{2}}$, and everything would be fine, but, as we just mentioned, $\Lambda_{\frac{1}{2}}$ is not unitarity because the generators are not Hermitian.

- Under an infinitesimal Lorentz transformation parametrized by $\omega_{\mu\nu}$, we have $\overline{\psi} \to \psi^\dagger \left(1 - \frac{i}{2}\omega_{\mu\nu}(S^{\mu\nu})^\dagger\right)\gamma^0$. The sum over μ and ν has six distinct nonzero terms. We have the transformation law $\overline{\psi} \to \psi\Lambda_{\frac{1}{2}}^{-1}$. Therefore, $\overline{\psi}\psi$ is a Lorentz scalar. Similarly, $\overline{\psi}\gamma^\mu\psi$ is a Lorentz vector.

- A Lorentz transformation acts on Dirac spinors via multiplication by a 4×4, complex matrix M

$$\psi' = M \cdot \psi = \begin{pmatrix} S & 0 \\ 0 & (S^{-1})^\dagger \end{pmatrix} \cdot \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

  where $S \in SL(2, \mathbb{C})$.

3.4 Spinor fields, Dirac equation

- A *spinor field* assigns a spinor to every point in spacetime. These fields describes fermions (i.e. particles of half-spin).

- The field equation for a *Dirac field* $\psi(x)$ (i.e. a field that transforms as a Dirac spinor under Lorentz transformations) is the *Dirac equation*

$$(i\partial\!\!\!/ - m)\psi = 0$$

  This implies the Klein-Gordon equation.

- In terms of the Weyl spinors ψ_L, ψ_R, the Dirac equation is

$$\begin{pmatrix} -m & i(\partial_0 + \vec{\sigma} \cdot \vec{\nabla}) \\ i(\partial_0 - \vec{\sigma} \cdot \vec{\nabla}) & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0$$

- In the limit $m \to 0$, the equations for ψ_L, ψ_R decouple, yielding the *Weyl equations*:

$$i(\partial_0 - \vec{\sigma} \cdot \vec{\nabla})\psi_L = 0$$
$$i(\partial_0 + \vec{\sigma} \cdot \vec{\nabla})\psi_R = 0$$

- The Lorentz-invariant Dirac Lagrangian is $\mathcal{L}_{Dirac} = \overline{\psi}(i\partial\!\!\!/ - m)\psi$. The Euler-Lagrange equation for $\overline{\psi}$ (or $\psi^\dagger$) immediately yields the Dirac equation.

3.5 Quantization of spinor fields (superstuff)

See the appendix for more information on superanalysis.

- Components of bosonic fields (scalar fields and fields with a 4-vector index) take values in $\mathbb{C}_c$; that is $\phi(x)$ at each location x of spacetime is a commuting quantity, possibly subject to an additional reality constraint if $\phi(x)$ is a real scalar field.

- Components of spinor fields take values in $\mathbb{C}_a$. That is, $\psi_L^1(x)$ and $\psi_L^2(x)$ at each spacetime location x are both a-numbers and so anticommute with each other and with components $\psi_L^1(x')$ and $\psi^2(x')$ at other spacetime locations x' and also with components of other spinor fields.

- Lagrangian density takes values in the real c-numbers $\mathbb{R}_c$.

- To resolve the ambiguity of ordering anticommuting quantities, we adopt the following normal-ordering scheme:
  - Amplitudes of positive-frequency waves are written without a star.
  - Amplitudes of negative-frequency waves are written with a star.
  - In expressions containing both starred and unstarred amplitudes, the starred amplitudes are written to the left and the unstarred amplitudes are written to the right (e.g. $\mathcal{A}^*\mathcal{A}$ is normal-ordered).

The purpose of this scheme is that, after promotion of amplitudes to quantum operators and takin expectation values, normal ordering results in a non-negative real number for all physical states ($\langle \widehat{\mathcal{A}}^\dagger \widehat{\mathcal{A}} \rangle \geq 0$).

- Accounting for the a-number property of components of spinor fields, fermion biolinear terms in the Lagrangian density can be rearranged using the identity $\chi^T A \psi = -\psi^T A^T \chi$. For instance, $u_L^\dagger \overline{\sigma^\mu} d_L = -(d_L)^T (\overline{\sigma^\mu})^T (u_L)^*$ for two left-handed Weyl spinors u_L and d_L.

4 Gauge theories

4.1 Electromagnetism ($U(1)$ gauge symmetries and Maxwell's equations)

- The Lagrangian density for a complex Dirac field is symmetric under the transformation $\psi' = e^{-iq\theta}\psi$, and as a result has a conserved current $j^\mu = q\overline{\psi}\gamma^\mu\psi$. The charge associated with this current is $Q = q(N_a - N_b)$ where N_a and N_b are the numbers of the Dirac particles and antiparticles, respectively.

- A *gauge transformation* is any transformation of a quantity in a physics equation that has no physical consequences, such as $\psi' = e^{-iq\theta}\psi$.

- The set of transformation operators $M(\theta) = e^{-iq\theta}$, where θ is a continuous parameter, form a group under simple multiplication. Namely, they form $U(1)$, the set of unitary ($M^\dagger M = I$) 1×1 matrices.

- $\psi' = e^{-iq\theta}\psi$ is a *global gauge transformation* because the phase change is applied universally to the field ψ independent of position. We can make this $U(1)$ transformation a *local gauge transformation* by allowing the parameter θ to depend on position.

- Under a local $U(1)$ gauge transformation, the Lagrangian density is not invariant. However, if we add a new (real) field $A_\mu(x)$, with
$$\mathcal{L} = i\overline{\psi}\partial\!\!\!/\psi - m\overline{\psi}\psi - j^\mu A_\mu$$
where $j^\mu = q\overline{\psi}\gamma^\mu\psi$, then the Lagrangian density will be invariant under the gauge transformation so long as we require $A'_\mu = A_\mu + \partial_\mu\theta(x)$ under the same transformation.

- We call $\vec{A}$ the *magnetic vector potential* and A^μ the *four-potential*.

- Define a *covariant derivative*[10] $D_\mu\psi = (\partial_\mu + iqA_\mu\psi)$. $D_\mu\psi$ transforms under the local transformation just like $\psi_\mu\psi$ does under the global transformation; we can rewrite the above Lagrangian density as
$$\mathcal{L} = i\overline{\psi}D\!\!\!\!/\,\psi - m\overline{\psi}\psi$$

- We need terms in the Lagrangian density that describe a free field; the first two terms do this for ψ, but we need the analogous for A_μ. Define the *Faraday tensor* $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Conventionally normalizing A_μ, the complete Lagrangian density becomes
$$\mathcal{L} = i\overline{\psi}\partial\!\!\!/\psi - m\overline{\psi}\psi - j^\mu A_\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$
The free-field Lagrangian density for the field represented by A_μ is therefore $\mathcal{L}_{A,free} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$.

- The components for the $F_{\mu\nu}$ tensor are
$$F_{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix}$$
For $F^{\mu\nu}$,
$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix}$$
Compactly, $F^{0i} = -F_{0i} = -E^i$ and $F^{ij} = F_{ij} = -\epsilon_{ijk}B^k$.

[10]One might recognize the covariant derivative from general relativity; both work with the notion of a connection, but general relativity works with the Levi-Civita connection over the tangent bundle of a pseudo-Riemannian manifold, while our connection is over the gauge bundle.

- By definition, we have the *Bianchi identity*

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0$$

- This stands for 64 equations, but most of these equations are identically zero; the four meaningful equations that emerge are the components of the source-free *Maxwell's equations* in differential form,

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

- The Euler-Lagrange equations applied to the full Lagrangian density imply that $\partial_\mu F^{\mu\nu} = j^\nu$ which become the other two Maxwell's equations

$$\vec{\nabla} \cdot \vec{E} = \rho, \quad \vec{\nabla} \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \vec{j}$$

- $F^{\mu\nu}$ is antisymmetric by construction, so

$$\partial_\nu \partial_\mu F^{\mu\nu} = \partial_\mu \partial_\nu F^{\mu\nu} = -\partial_\mu \partial_\nu F^{\nu\mu} = -\partial_\nu \partial_\mu F^{\mu\nu} = 0$$

meaning that, since $\partial_\mu F^{\mu\nu} = j^\nu$, we have

$$0 = \partial_\nu(\partial_\mu F^{\mu\nu}) = \partial_\nu j^\nu$$

- It is convenient to choose a gauge (that is, choose a specific function $\theta(x)$ that makes working with A^μ easier. For example, the *Lorenz gauge* is the gauge for which $\partial_\mu A^\mu = 0$. This is handy because the Euler-Lagrange equations for the field become

$$\Box A^\mu - j^\mu = 0$$

In the absence of a current, the equations are $\Box A^\mu = 0$, which are simply four independent Klein-Gordon equations for a massless field.

- The *Coulomb gauge* is that in which the Lorenz gauge condition is satisfied and additionally $\vec{\nabla} \cdot \vec{A} = 0$. We can require that in free space, $A^t = 0$, making the free field's stress-energy tensor easier.

4.2 Massive 4-vector fields

- One can modify electrodynamics by including a non-zero mass m for a 4-vector field $Z^\mu(x)$; the kinetic term then takes the modified form

$$\mathcal{L} = -\frac{1}{4} Z_{\mu\nu} Z^{\mu\nu} + \frac{1}{2} m^2 Z^\mu Z_\mu$$

where $Z_{\mu\nu} = \partial_\mu Z_\nu - \partial_\nu Z_\mu$ is the usual field strength.

- The equations of motion are $(\Box + m^2) Z^\mu = 0$, $\partial_\mu Z^\mu = 0$.

4.3 Nonabelian gauge theories

- In general, a gauge theory is defined through the following ingredients:
  - A Lie group G, the *gauge group*, and its Lie algebra $\mathfrak{g}$.
  - A coupling strength g_C for each simple factor in G; in the following, we take G to be simple, so there is just one coupling strength.
  - For each matter field, a Lie algebra representation $r : \mathfrak{g} \to \mathfrak{h}$ mapping each generator X of $\mathfrak{g}$ to an $n \times n$ matrix $r(X)$.

We demand that the gauge group be compact, so that gauge fields have proper sign kinetic terms, and that the representation matrices $\mathfrak{h}$ be a subalgebra of the Hermitian matrices. These ingredients alone determine the set of gauge fields, matter fields, their kinetic terms, and the interaction terms coupling gauge fields to matter fields. One can optionally then include extra G-invariant terms in the Lagrangian coupling matter fields to each other.

- For each basis element X_I of $\mathfrak{g}$, there is one 4-vector *gauge field* $G^I_\mu(x)$, $I = 1, \cdots, \dim G$. We can optionally package these into a single matrix-valued gauge field

  $$G_\mu(x) = G^I_\mu(x) X_I \in \mathfrak{g}$$

  for each value of μ, x. When the Lie algebra index I is missing, this matrix-valued field is implied.

- The *nonabelian field strength tensor* is defined as

  $$G_{\mu\nu}(x) = \partial_\mu G_\nu - \partial_\nu G_\mu + ig_C[G_\mu, G_\mu]$$

  which evidently is $\mathfrak{g}$-valued for each value of μ, ν, x. (For $G = U(1)$, the commutator term vanishes and this reduces to the definition of the electromagnetic field strength $F_{\mu\nu}$)

- Because $G_{\mu\nu}$ is $\mathfrak{g}$-valued, to put it in the Lagrangian, you need to make it into a scalar, so you take the trace: $\mathcal{L} = -\frac{1}{4}\mathrm{tr}(\mathrm{G}^{\mu\nu}\mathrm{G}_{\mu\nu})$. Some authors, e.g. Schwartz, write instead $-\frac{1}{4}G^2_{\mu\nu}$, since $G^{\mu\nu}G_{\mu\nu} = G^{I\ \mu\nu}G^J_{\mu\nu}X^I X^J$, so $\mathrm{tr}(\mathrm{G}^{\mu\nu}\mathrm{G}_{\mu\nu}) = \mathrm{G}^{\mathrm{I}\ \mu\nu}\mathrm{G}^{\mathrm{J}}_{\mu\nu}\,\mathrm{tr}(\mathrm{X^I X^J}) \equiv \mathrm{G}^2$ up to a normalization factor. Both notations show up in these notes.

- A *gauge transformation* is a choice of $g(x) = e^{i\epsilon^I(x)X_I} \in G$ that could vary smoothly from one location x to another. This gauge transformation acts on gauge fields as

  $$\begin{aligned} G'_\mu &= gG_\mu g^{-1} - \frac{i}{g_C}(\partial_\mu g)g^{-1} \\ \implies G'_{\mu\nu} &= gG_{\mu\nu}g^{-1} \end{aligned}$$

- The matter fields are a multiplet of n fields $\psi(x) = (\psi_1(x), \cdots, \psi_n(x))^T$ that will be acted on by the gauge transformation. Namely, the Lie algebra representation $r : \mathfrak{g} \to \mathfrak{h}$ induces a Lie group representation $R : G \to H$ by exponentiating the $n \times n$ matrices, with the understanding that R could be projective. We have

  $$\psi'(x) = R(g(x))\psi(x) = e^{i\epsilon^I(x)r(X_I)}\psi(x)$$

- The nonabelian covariant derivative acting on the multiplet $\psi(x)$ is

  $$D_\mu\psi = \partial_\mu\psi - ig_C G^I_\mu r(X_I)\psi$$

  This behaves nicely under gauge transformations:

  $$D'_\mu\psi' = \partial_\mu\psi' - ig_C G^{I'}_\mu r(X_I)\psi' = R(g(x))D_\mu\psi$$

- Using the nonabelian field strength and the covariant derivative acting on the matter field, both of which objects transform nicely under gauge transformations, the kinetic and gauge-matter interaction terms in the Lagrangian density are

  $$\mathcal{L} = -\frac{1}{4}\mathrm{Tr}(G^{\mu\nu}G_{\mu\nu}) + \overline{\psi}(i\not{D} - m)\psi$$

  where Tr indicates trace of the gauge indices. Gauge invariance of this expression follows from the cyclical property of trace and unitarity of the matrices $R(g(x))$.

5 Spontaneous symmetry breaking

- The distinction between *spontaneous* and *explicit symmetry breaking* is that with spontaneous symmetry breaking the Lagrangian is invariant under the symmetry, but the ground state of the theory is not. WIth explicit symmetry breaking, there was never an exact symmetry to begin with. One usually associates spontaneous symmetry breaking with phase transitions. One can say a tremendous amount about the broken phase with an effective field theory with an effective field theory whose only input is the symmetry that was broken.

- An example of spontaneous symmetry breaking is in the context of ferromagnetic materials, such as iron. The magnetic moment of such a material can be represented by a field $\vec{M}(x)$ related to the local direction the spins are pointing. At high temperature, the entropic term in the free energy, $F = E - TS$, dominates the energetic one and $\vec{M}(x)$ points in random directions at each x. When a ferromagnetic material is cooled below its Curie temperature, the energetic contribution to free energy, which is lower when neighboring atoms are aligned, starts to dominate; domains with aligned spins start to grow, and long-range order emerges. We write $\vec{M}(x) = \vec{\mu} + \vec{\sigma}(x)$ where $\vec{\mu}$ is the expectation value of $\vec{M}$ in the vacuum and $\vec{\sigma}$ are local excitations around this minimum. At high temperature, the theory has a rotational symmetry (no direction is distinguished), but at low temperature, this symmetry is spontaneously broken, since $\vec{\mu}$ points in some particular direction. The field $\vec{\sigma}(x)$ that encodes excitations around the vacuum encodes spin waves whose quanta are a special example of Goldstone bosons.

- *Goldstone's theorem* states that spontaneous breaking of continuous global symmetries implies the existence of massless particles, called *Goldstone bosons*. If the symmetry is gauged, then in the broken phase the gauge boson will acquire a mass.

5.1 The sigma model

- As an example, consider a complex scalar field with Lagrangian

$$\mathcal{L} = (\partial_\mu \phi^*)(\partial_\mu \phi) + m^2 \phi\phi^* - \frac{\lambda}{4}\phi^2\phi^{*2}$$

  This theory has a global $U(1)$ symmetry.

- When a constant value $\phi = v$ satisfies the classical equations of motion, the classical expectation is that the field should be at this value over all space. We can identify this classical expectation with a quantum *vacuum expectation value* $v = \langle\Omega|\phi|\Omega\rangle$.

- For $m^2 > 0$, our theory is unstable around $\phi = 0$. The potential $V(\phi) = -m^2|\phi|^2 + \frac{\lambda}{4}|\phi|^4$ is minimized when $|\phi|^2 = \frac{2m^2}{\lambda}$, so there are an infinite number of equivalent vacua $|\Omega_\theta\rangle$ with $\langle\Omega_\theta|\phi|\Omega_\theta\rangle = \sqrt{\frac{2m^2}{\lambda}}e^{i\theta}$ for any constant θ. All the vacua are equivalent by symmetric, so we conventionally pick $|\Omega\rangle$ s.t. $\langle\Omega|\phi|\Omega\rangle = v = \sqrt{\frac{2m^2}{\lambda}}$.

- Expanding aorund v by parametrizing $\phi(x)$ in terms of two real fields $\sigma(x), \pi(x)$ as

$$\phi(x) = \left(\sqrt{\frac{2m^2}{\lambda}} + \frac{1}{\sqrt{2}}\sigma(x)\right) e^{i\frac{\pi(x)}{F_\pi}}$$

  for some $F_\pi \in \mathbb{R}$, then $V(\phi)$ depends only on σ. Expanding the Lagrangian around the minimum,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\sigma)^2 + \left(\sqrt{\frac{2m^2}{\lambda}} + \frac{1}{\sqrt{2}}\sigma(x)\right)^2 \frac{1}{F_\pi^2}(\partial_\mu\pi)^2 - \left(-\frac{m^4}{\lambda} + m^2\sigma^2 + \frac{1}{2}\sqrt{\lambda}m\sigma^3 + \frac{1}{16}\lambda\sigma^4\right)$$

  Choosing $F_\pi = \sqrt{2}v$ makes the π kinetic term canonically normalized. This theory is called a *linear sigma model*. The π field is massless and is the Goldstone boson; it is often called a pion because it is closely related to the real-world hadrons $\pi^\pm, \pi^0$.

- In the linear sigma model, the σ field has mass $m_\sigma = \sqrt{2}m$; it can be visualized as radial excitations of the "Mexican hat potential".

- Goldstone bosons are naturally associated with *shift symmetries*. The vacuum $\langle\phi\rangle = \sqrt{\frac{2m^2}{\lambda}}$ breaks the symmetry $\phi(x) \to e^{i\theta}\phi(x)$. The symmetry is still realized as $\pi(x) \to \pi(x) + F_\pi\theta$ with σ invariant. In particular, the shift symmetry forbids a mass term for $\pi(x)$.

- To distinguish the Goldstone bosons, whose interactions are determined by symmetry, from the radial modes (such as σ) which are model dependent and invariant under the symmetry, we can take the limit $m, \lambda \to \infty$ keeping $F_\pi = \frac{2m}{\sqrt{\lambda}}$ fixed, then the Lagrangian reduces to $\mathcal{L} = \frac{1}{2}(\partial_\mu\pi)^2$, which is a theory of a free pion. This *decoupling limit* is much more interesting in theories where pions are not free particles. This Lagrangian is an example of a *nonlinear sigma model*, which is the linear sigma model in which the σ field has been decoupled.

5.2 The abelian Higgs model

- We have seen that a spontaneously broken continous global symmetry generates Goldstone bosons transforming as elements of the coset G/H, where G is the original symmetry group and H is the symmetry group of the vacuum. Now we consider what happens if there is a gauge boson associated with the broken symmetry. As we will see, this causes the Goldstone boson to disappear from the spectrum and the gauge boson to become massive through a procedure known as the Higgs mechanism.

- Let us return to the linear sigma model and gauge the $U(1)$ symmetry; the Lagrangian is then

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + (\partial_\mu\phi^* - ieA_\mu\phi^*)(\partial_\mu\phi + ieA_\mu\phi) + m^2|\phi|^2 - \frac{\lambda}{4}|\phi|^4$$

  which is known as the *abelian Higgs model*. The wrong-sign mass term for the scalar indicates that the ground state has $|\langle\phi\rangle| = \frac{v}{\sqrt{2}} = \sqrt{\frac{2m^2}{\lambda}}$.

- To see what happens to them, we write $\phi(x) = \left(\frac{v+\sigma(x)}{\sqrt{2}}\right)e^{i\frac{\pi(x)}{F_\pi}}\phi(x)$. Plugging this in and looking at terms involving only A_μ, $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \frac{1}{2}e^2v^2A_\mu^2 + \cdots$, suggesting that the gauge moson has picked up a mass $m_A = ev$. Similarly, the σ field has mass $m_\sigma = \sqrt{2}m$ and π is massless. But because there are bilinear terms miaxing π, σ, A_μ, extracting the spectrum is not that simple.

- Decoupling σ through the limit $m, \lambda \to \infty$ with v fixed, the Lagranigan simplifies to $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \frac{1}{2}m_A^2\left(A_\mu + \frac{1}{eF_\pi}\partial_\mu\pi\right)^2$, implying that we should set $F_\pi = v$ so that $\pi(x)$ has canonical normalization. This Lagrangian has a gauge boson mass term, a kinetic term for π, as well as an $A_\mu\partial_\mu\pi$ cross term indicating mixing between π and A_μ.

- The gauge symmetry of the above Lagrangian is $A_\mu(x) \to A_\mu(x) + \frac{1}{e}\partial_\mu\alpha(x)$, $\pi(x) \to \pi(x) - F_\pi\alpha(x)$. Now we can just remove the kinetic mixing by choosing a gauge. In the *unitary gauge*, we use the shift to set $\pi(x) = 0$. In this gauge, the Lagrangian becomes simply that of a massive gauge boson. Another convenient gauge is the Lorenz gauge, which makes the cross term vanish, and the π field is massless with a normal kinetic term with the correct sign. In this gauge, the constrained gauge field has two degrees of freedom and the pion has one degree of freedom, which are the same three degrees of freedom of the unconstrained massive gauge boson in the unitary gauge. Thus, we say that, in the unitary gauge, the gauge boson eats the Goldston boson through the *Higgs mechanism*.

- In the case of broken local symmetries, the low-energy theory has no memory that the symmetry was spontaneously broken instead of explicitly broken. With explicit symmetry breaking we can turn the abelian Higgs model Lagrangian into the nonlinear sigma model by integrating a pion, i.e. performing a field redefinition $A_\mu \to A_\mu + \frac{1}{eF_\pi}\partial_\mu$, introducing a gauge invariance, which can be used to set $\pi = 0$, reverting to the theory with the massive gauge boson. Introducing pions in this way is an efficient way to study the high-energy properties of a theory with a massive gauge boson, since scalars are easier to compute with than longitudinal modes of gauge bosons.

- Spontaneously broken theories are renormalizable while explicitly broken ones are not. The difference between the two is the σ field, present in the spontaneously broken theory, but not in the explicitly broken one.
- The abelian Higgs model is a special case that happens to be renormalizable without the σ field because the photon has no self-interactions, or equivalently, because the π field has no interactions in teh nonlinear sigma model.

5.3 Weak interactions as an $SU(2)$ Yang-Mills theory

Note that what follows is just a "test theory"–the relevant symmetry for the standard model is $SU(2)_L \times U(1)$ (a unified theory describing an "electroweak interaction").

- Consider a doublet of Dirac spinors $\psi = \binom{\psi_{\nu_e}}{\psi_e}$ and $\overline{\psi} = [\overline{\psi}_{\nu_e}\ \overline{\psi}_e]$ where each entry in this doublet is itself a four-component Dirac field, one for the electron and one for the electron neutrino. One usually replaces the ψ_is with $e = \psi_e$, $\nu_e = \psi_{\nu_e}$, $\overline{e} = \overline{\psi}_e$, $\overline{\nu}_e = \overline{\psi}_{\nu_e}$ (note that the last is not the field for an antineutrino).
- $\overline{\psi}\psi$ is symmetric under a global $SU(2)$ transformation.
- A weak-interaction Lagrangian density that preserves the magnitude of $\overline{\psi}\psi = \overline{\nu}_e\nu_e + \overline{e}e$ for rotations of e and ν_e fields would imply that these fields are identical with respect to the weak interaction. We are not talking about physical rotations, but rather matrices M such that $\psi' = M\psi$ mix the component fields of ψ without affecting the value of $\overline{\psi}\psi$; this group turns out to be $SU(2)$. From the $SU(2)$ symmetry we get the conserved currents

$$\begin{aligned} J^{(x)\mu} &= \frac{1}{2}g_W(\overline{\nu}_e\gamma^\mu e + \overline{e}\gamma^\mu\nu_e) \\ J^{(y)\mu} &= -\frac{1}{2}g_W(\overline{\nu}_e\gamma^\mu e - \overline{e}\gamma^\mu\nu_e) \\ J^{(z)\mu} &= \frac{1}{2}g_W(\overline{\nu}_e\gamma^\mu\nu_e - \overline{e}\gamma^\mu e) \end{aligned}$$

  with some constant g_W, where the x, y, z superscripts are in parentheses to emphasize that these are just indices for the three currents, not components of the field (which is what the μ is for). We package the three into a single vector $\vec{J}^\mu$ (we reuse this notation in what follows).
- For local $SU(2)$ symmetry, introduce (3) new fields $\vec{W}_\mu$ and define a covariant derivative $D_\mu = \partial_\mu + i\frac{1}{2}g_W\vec{\sigma}\cdot\vec{W}_\mu$. The field strength tensor is

$$\vec{F}_{\mu\nu} = \partial_\mu\vec{W}_\nu - \partial_\nu\vec{W}_\mu - g_W\vec{W}_\mu \times \vec{W}_\nu$$

  We get a Lagrangian density

$$\mathcal{L} = -\frac{1}{4}\vec{F}_{\mu\nu}\cdot\vec{F}^{\mu\nu} + i\overline{\psi}\not{D}\psi = i\overline{\psi}\not{\partial}\psi - \vec{J}^\mu\cdot\vec{W}_\mu$$

  The three new fields $\vec{W}_\mu$ correspond to the *vector bosons*. Since we have no mass terms for these fields (we cannot have them because we cannot make them gauge-invariant), these vector bosons should be massless, in direct contradiction to experimental evidence. We readdress this later.
- Define $J^{+\mu} = g_W\overline{\nu}_e\gamma^\mu e$, $J^{-\mu} = g_W\overline{e}\gamma^\mu\nu_e$,[11] and $W^\pm_\mu = \sqrt{\frac{1}{2}}(W^{(x)}_\mu \mp iW^{(y)}_\mu)$, so that $\mathcal{L}_{int} = -\sqrt{\frac{1}{2}}(J^{+\mu}W^+_\mu + J^{-\mu}W^-_\mu) - J^{(z)\mu}W^{(z)}_\mu$. Now, each interaction term describes a vertex in a Feynman diagram! $J^{+\mu}W^+_\mu$

[11] In ordinary quantum mechanics we define the angular momentum raising and lowering operators $J_\pm = J_x \pm iJ_y$; because the $SU(2)$ symmetry obeyed here is analogous to the $SU(2)$ symmetry that describes spin states, we reapply the same kind of trick, defining $J^{+\mu} = J^{(x)\mu} + iJ^{(y)\mu}$, $J^{-\mu} = J^{(x)\mu} - iJ^{(y)\mu}$.

describes a vertex where an electron and the vector boson corresponding to the W^+_μ field enter and a neutrino leaves; since the electron has a negative charge and the neutrino has no charge, the boson corresponding to the W^+_μ field has a positive charge. Similarly, $J^{-\mu}W^-_\mu$ describes a vertex $\nu + W^- \to e$. The last term describes a pair of vertices where either a neutrino or electron enters and departs unchanged; the boson associated with the $W^{(z)}$ field therefore has zero charge. Note that the $W^\pm_\mu$ fields are adjoints of each other, so their corresponding bosons are antiparticles of each other, whereas the $W^{(z)}$ field is real and therefore the corresponding particle is its own antiparticle.

- Looking ahead, electroweak interactions are a nonabelian gauge theory with gauge group $SU(2) \times U(1)$ with the left-handed quarks and leptons transforming as fundamental 2×2 representations, the right-handed quarks and leptons transforming trivially, and the Higgs scalar field transforming in the 2×2 representation of the $SU(2)$ factor.

- The weak interaction only operates between particles with left-handed chirality, as such, the local gauge symmetry is not written $SU(2)$, but rather $SU(2)_L$, a symmetry that does not apply to chirally right-handed particles. This must be accounted for in writing the currents; we make this correction, and others–i.e., the fact that it does not deal with interactions involving the Z vector boson, and why we have not accounted for the mass of the $W^\pm$ and Z bosons–in what follows.

5.4 Electroweak symmetry breaking

- Electroweak unification is based on the symmetry breaking of $SU(2) \times U(1)_Y \to U(1)_{\rm EM}$. The high-energy $U(1)$ symmetry is called the *hypercharge*[12], denoted $U(1)_Y$. The photon is a linear combination of the hypercharge gauge boson and one of the generators of $SU(2)$.

- In the standard model, $SU(2) \times U(1)_Y$ is broken by the *vacuum expectation value (vev)* of a complex doublet H with hypercharge $\frac{1}{2}$ called the *Higgs multiplet*. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}(W^a_{\mu\nu})^2 - \frac{1}{4}B^2_{\mu\nu} + (D_\mu H)^\dagger (D_\mu H) + m^2 H^\dagger H - \lambda (H^\dagger H)^2$$

where B_μ is the hypercharge gauge boson, with $B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$, and where W^a_μ are the $SU(2)$ gauge bosons, with $W^a_{\mu\nu}$ their field strengths. The normalization of the $\lambda(H^\dagger H)^2$ term is conventional. The covariant derivative is

$$D_\mu H = \partial_\mu H - igW^a_\mu \tau^a H - \frac{1}{2} i g' B_\mu H$$

with $\tau^a = \frac{1}{2}\sigma^a$ the canonically normalized $SU(2)$ generators, and where g, g' are the $SU(2), U(1)_Y$ couplings,respectively.

- The Higgs potential $V(H) = -m^2|H|^2 + \lambda|H|^4$ introduces a vev for H, which we can take to be real and in the lower component without loss of generality. Thus we can expand

$$H = \exp\left(2i\frac{\pi^a \tau^a}{v}\right)\begin{pmatrix} 0 \\ \frac{v+h}{\sqrt{2}} \end{pmatrix}$$

with $v = \frac{m}{\sqrt{\lambda}}$. The factors of $\frac{1}{\sqrt{2}}$ convert between the canonical normalization of a complex scalar (H) and a real scalar (h).

- Plugging in the vev,

$$|D_\mu H|^2 = g^2 \frac{v^2}{8}\left[(W^1_\mu)^2 + (W^2_\mu)^2 + \left(\frac{g'}{g}B_\mu - W^3_\mu\right)^2\right]$$

These are the mass terms associated with three masive gauge bosons.

[12] Not to be confused with the low-energy $U(1)$ associated with electromagnetism, denoted $U(1)_{\rm EM}$.

- To diagonalize the masses, we note that, because the kinetic terms for B_μ and W^a_μ are canonically normalized, we should only rotate and not rescale the gauge bosons in the diagonalization. Thus we redefine

$$\begin{aligned} Z_\mu &\equiv \cos\theta_w\, W^3_\mu - \sin\theta_w\, B_\mu \\ A_\mu &\equiv \sin\theta_w\, W^3_\mu + \cos\theta_w\, B_\mu \end{aligned} \qquad \Longleftrightarrow \qquad \begin{aligned} B_\mu &= \cos\theta_w\, A_\mu - \sin\theta_w\, Z_\mu \\ W^3_\mu &= \sin\theta_w\, A_\mu + \cos\theta_w\, Z_\mu \end{aligned}$$

  with the rotation angle defined by $\tan\theta_w = \frac{g'}{g}$.

- With these definitions, the kinetic terms in the Lagrangian for Z_μ and the photon A_μ are

$$\mathcal{L} = -\frac{1}{4}F^2_{\mu\nu} - \frac{1}{4}Z^2_{\mu\nu} + \frac{1}{2}m_Z^2 Z^\mu Z_\mu$$

  with $m_Z = \frac{1}{2\cos\theta_w}gv$, $Z_{\mu\nu} = \partial_\mu Z_\nu - \partial_\nu Z_\mu$, and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

- Gauge bosons transform in the adjoint representation, so their interactions are determined by commutators. In particular, since the photon is part of W^3_μ, its couplings to the W^a_μ are determined by $g[A_\mu, W^a\tau^a] = g\sin\theta_w W^3 W^a[\tau^3,\tau^a]$. Thus, the electromagnetic coupling strength is set by $e = g\sin\theta_w = g'\cos\theta_w$.

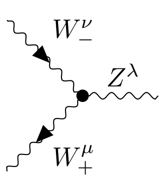
- Define $\tau^\pm = \frac{1}{\sqrt{2}}(\tau^1 \pm i\tau^2)$. Since $[\tau^3,\tau^\pm] = \pm\tau^\pm$, the W-boson that couples to $\tau^\pm$ has charge ±1. We write $\tau^a W^a = W^+\tau^+ + W^-\tau^- + W^3\tau^3$, i.e. $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu)^2$.

- The gauge Lagrangian is

$$\begin{aligned} \mathcal{L}_{\text{gauge}} = -\frac{1}{4}F^2_{\mu\nu} - \frac{1}{4}Z^2_{\mu\nu} + \frac{1}{2}m_Z^2 Z^\mu Z_\mu - \frac{1}{2}\left(\partial_\mu W^+_\nu - \partial_\nu W^+_\mu\right)\left(\partial_\mu W^-_\nu - \partial_\nu W^-_\mu\right) + m_W^2 W^+_\mu W^{-\mu} \\ -ie\cot\theta_w\Big[\partial_\mu Z_\nu\left(W^+_\mu W^-_\nu - W^+_\nu W^-_\mu\right) + Z_\nu\left(-W^+_\mu\partial_\nu W^-_\mu + W^-_\mu\partial_\nu W^+_\mu + W^+_\mu\partial_\mu W^-_\nu - W^-_\mu\partial_\mu W^+_\nu\right)\Big] \\ -ie\Big[\partial_\mu A_\nu\left(W^+_\mu W^-_\nu - W^+_\nu W^-_\mu\right) + A_\nu\left(-W^+_\mu\partial_\nu W^-_\mu + W^-_\mu\partial_\nu W^+_\mu + W^+_\mu\partial_\mu W^-_\nu - W^-_\mu\partial_\mu W^+_\nu\right)\Big] \\ -\frac{1}{2}\frac{e^2}{\sin^2\theta_w}W^+_\mu W^{+\mu}W^-_\nu W^{-\nu} + \frac{1}{2}\frac{e^2}{\sin^2\theta_w}W^+_\mu W^-_\nu W^{+\mu}W^{-\nu} \\ -e^2\cot^2\theta_w\left(Z_\mu W^{+\mu}Z_\nu W^{-\nu} - Z_\mu Z^\mu W^+_\nu W^{-\nu}\right) + e^2\left(A_\mu W^{+\mu}A_\nu W^{-\nu} - A_\mu A^\mu W^+_\nu W^{-\nu}\right) \\ +e^2\cot\theta_w\Big[A_\mu W^{+\mu}W^-_\nu Z^\nu + A_\mu W^-_\mu Z_\nu W^{+\nu} - W^+_\mu W^{-\mu}A_\nu Z^\nu\Big] \end{aligned}$$

  with $m_W = \frac{v}{2}g$, $m_Z = \frac{1}{2\cos\theta_w}gv = \frac{v}{2}\sqrt{g^2+g'^2} = \frac{m_W}{\cos\theta_w}$, which tells us that the $W^\pm$ bosons should be lighter than the Z boson.

- The Feynman rules can be read off this Lagrangian. For example, the vertex $W^\mu_+(p_1)W^\nu_-(p_2)Z^\lambda(p_3)$ with all the momenta incoming is



$$= -ie\cot\theta_w\left[g^{\mu\nu}(p_1-p_2)^\lambda + g^{\nu\lambda}(p_2-p_3)^\mu + g^{\lambda\mu}(p_3-p_1)^\nu\right]$$

  We take W^- to be the particle and W^+ to be its antiparticle, so we draw paticle-flow arrows in the direction of momentum for W^- and opposite to the direction of momentum for W^+.

- The field h, known as the *Higgs boson*, is present even in the unitary gauge. While the Higgs doublet H has charges under the weak and hypercharge gauge groups, the Higgs boson h does not. Expanding out the Lagrangian, we find that the terms involving the Higgs boson are

$$\begin{aligned}\mathcal{L}_{\text{Higgs}} &= -\frac{1}{2}h\left(\Box + m_h^2\right)h - g\frac{m_h^2}{4m_W}h^3 - \frac{g^2}{32}\frac{m_h^2}{m_W^2}h^4 + 2\frac{h}{v}\left(m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu\right) \\ &+ \left(\frac{h}{v}\right)^2\left(m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu\right)\end{aligned}$$

where $m_h = \sqrt{2}m$.

6 The standard model

6.1 Coupling electroweak gauge bosons to fermions

- When we viewed weak interactions as an $SU(2)$ Yang-Mills theory, we noted that the weak interaction only operates between left-handed particles; in the language of gauge bosons, we say that $SU(2)$ gauge bosons only couple to left-handed fermions. The left-handed leptons transform under $SU(2)$ in the fundamental representation, as do the left-handed quarks.
- The three *generations* of $SU(2)$ doublet pairs of quarks and leptons can be written
$$L^i = \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix}, \begin{pmatrix} \nu_{\mu L} \\ \mu_L \end{pmatrix}, \begin{pmatrix} \nu_{\tau L} \\ \tau_L \end{pmatrix}, \quad Q^i = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \begin{pmatrix} c_L \\ s_L \end{pmatrix}, \begin{pmatrix} t_L \\ b_L \end{pmatrix}$$
where $i = 1, 2, 3$ indexes the generation. These all transform as left-handed Weyl spinors. We index right-handed fermions by the first-generation label:
$$\begin{aligned} e_R^i &= \{e_R, \mu_R, \tau_R\}, \quad \nu_R^i = \{\nu_{eR}, \nu_{\mu R}, \nu_{\tau R}\}, \\ u_R^i &= \{u_R, c_R, t_R\}, \quad d_R^i = \{d_R, s_R, b_R\} \end{aligned}$$
These all happen to be $SU(2)$ singlets so they are undcharged under the weak interaction. They transform as right-handed Weyl spinors under the Lorentz group.[13]
- All fermions couple to the hypercharge gauge boson. Writing Y_Q, Y_L for the left-handed fields' hypercharges (which happen to be the same for each generation) and Y_e, Y_ν, Y_u, Y_d for the right-handed fields' hypercharges (which also happen to be the same for each generation) and putting everything together, the gauge interactions are[14]
$$\begin{aligned} \mathcal{L} = {} & i\overline{L}_i \left(\partial\!\!\!/ - ig W\!\!\!\!/^{\,a} \tau^a - ig' Y_L B\!\!\!/\right) L_i + i\overline{Q}_i \left(\partial\!\!\!/ - ig W\!\!\!\!/^{\,a} \tau^a - ig' Y_Q B\!\!\!/\right) Q_i \\ & + i\overline{e}_R^i \left(\partial\!\!\!/ - ig' Y_e B\!\!\!/\right) e_R^i + i\overline{\nu}_R^i \left(\partial\!\!\!/ - ig' Y_\nu B\!\!\!/\right) \nu_R^i \\ & + i\overline{u}_R^i \left(\partial\!\!\!/ - ig' Y_u B\!\!\!/\right) u_R^i + i\overline{d}_R^i \left(\partial\!\!\!/ - ig' Y_d B\!\!\!/\right) d_R^i \end{aligned}$$
- Because hypercharge is a $U(1)$ group, the hypercharges could be arbitrary real numbers. To find the actual hypercharges in the standard model, use the known electric charges. First isolating the neutral gauge bosons, W^3 and B_μ, then changing to the $A_\mu - Z_\mu$ basis using the formula from the last chapter gives, for the electron and neutrino couplngs,
$$\begin{aligned} \mathcal{L} &= \overline{e}_L^i \left(-\frac{1}{2} g W\!\!\!\!/^{\,3} + g' Y_L B\!\!\!/\right) e_L^i + \overline{\nu}_L^i \left(\frac{1}{2} g W\!\!\!\!/^{\,3} + g Y_L B\!\!\!/\right) \nu_L^i \\ &\quad + g' Y_e \overline{e}_R^i B\!\!\!/ e_R^i + g' Y_\nu \overline{\nu}_R^i B\!\!\!/ \nu_R^i \\ &= e\left\{ \left(-\frac{1}{2} + Y_L\right) \overline{e}_L^i A\!\!\!/ e_L^i + \left(\frac{1}{2} + Y_L\right) \overline{\nu}_L^i A\!\!\!/ \nu_L^i + Y_e \overline{e}_R^i A\!\!\!/ e_R^i + Y_\nu \overline{\nu}_R^i A\!\!\!/ \nu_R^i \right\} + Z \text{ terms} \end{aligned}$$
Since the electric charges are the coefficients of the coupling to the photon, we can read off the relationship between hypercharges and electric charges. Using that the electron is conventionally defined to have charge -1, we see that $Y_L = -\frac{1}{2}$ and $Y_e = -1$, implying that ν_L is neutral, in agreement with nature, and for ν_R to be neutral we also need $Y_\nu = 0$. Similarly, using that the up quark has charge $\frac{2}{3}$ and down quark has charge $-\frac{1}{3}$, we need $Y_Q = \frac{1}{6}$, $Y_u = \frac{2}{3}$, and $Y_d = -\frac{1}{3}$.

[13] Right-handed neutrinos have not yet been observed, but many textbooks and treatments include them in case they do exist. As a fun coincidence, on the day of the lecture on this material, Quanta released an article that concluded that there is very strong evidence of their nonexistence.

[14] Since the fermions are all left- or right-handed Weyl spinors, it would be technically correct to replace $\overline{Q}_i \partial\!\!\!/ Q_i \to Q_i^\dagger \overline{\sigma} \partial_\mu Q_i$ and $\overline{u}_R^i \partial\!\!\!/ u_R^i \to u_R^{i\dagger} \sigma_\mu \partial u_R^i$. However, since we almost always compute in the broken phase, where left- and right-handed spinors combine into a single Dirac representation, it is generally easier to use Dirac-spinor notation from the start, where L, R indicate implicit chirality projectors. That is, $\overline{Q}_i \partial\!\!\!/ Q_i \to Q_i^\dagger \gamma^0 \partial\!\!\!/ P_L Q_i$ with $P_L = \frac{1}{2}(1 - \gamma^5)$ and $\overline{u}_R^i \partial\!\!\!/ u_R^i \to u_R^{i\dagger} \gamma^0 \partial\!\!\!/ P_R u_R^i$ with $P_R = \frac{1}{2}(1 + \gamma^5)$.

- A representation of $SU(2)_{\text{weak}} \times U(1)_Y$ has some matrix T^3 associated with W_μ^3 and a number Y associated with B_μ. In the fundamental representation of $SU(2)$, $T^3 = \tau^3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ but we can allow for more general possibilities. There are infinitely many represenations of $U(1)_Y$ since $\psi \to e^{i\alpha Y}\psi$ leaves $i\overline{\psi}(\not\partial - iYg'\not B)\psi$ invariant for any $Y \in \mathbb{R}$. Then, the part of the covariant derivative involving T^3 and Y is

$$\begin{aligned} D_\mu &= \partial_\mu - igW_\mu^3 T^4 - ig' B_\mu Y 1 \\ &= \partial_\mu - ieA_\mu(T^3 + Y1) - ieZ_\mu(\cos\theta_w T^3 - \tan\theta_w Y 1) \end{aligned}$$

where 1 is the identity matrix acting on the n-dimensional vector space on which T^3 acts in a given representation. So $Q = T^3 + Y1$ measures the electric charge; for example, $Q = \binom{0}{e_L} = (\tau^3 + Y_L 1_{2\times 2})\binom{0}{e_L} = -\binom{0}{e_L}$ shows that the left-handed electron has charge -1.

- We can write neutral gauge boson interactions with general fields ψ_L^j, ψ_R^j more generally as

$$\mathcal{L} = i\overline{\psi}_L^j(\not\partial - ig\not W^3 T^3 - ig'\not B Y_j^L)\psi_L^j + i\overline{\psi}_R^j(\not{+}ig'\not B Y_j^R)\psi_R^j$$

Then in terms of neutral currents that couple to the Z boson and photon we rewrite

$$\mathcal{L} = \cdots + \frac{e}{\sin\theta_w} Z_\mu J_\mu^Z + eA_\mu J_\mu^{\text{EM}}$$

with

$$J_\mu^Z = \cos\theta_w J_\mu^3 - \frac{\sin^2\theta_w}{\cos\theta_w} J_\mu^Y = \frac{1}{\cos\theta_w}(J_\mu^3 - \sin^2\theta_w J_\mu^{\text{EM}}$$

Assuming $SU(2)_{\text{weak}}$ acts only on left-handed states, the currents are

$$\begin{aligned} J_\mu^Y &= \sum_i Y_i^L \overline{\psi}_i^L \gamma^\mu \psi_i^L + \sum_i Y_i^R \overline{\psi}_i^R \gamma^\mu \psi_i^R = J_\mu^{\text{EM}} - J_\mu^3 \\ J_\mu^3 &= \sum_i \overline{\psi}_i^L \gamma^\mu T^3 \psi_i^L \end{aligned}$$

$$J_\mu^{\text{EM}} = \sum_i Q_i(\overline{\psi}_i^L \gamma^\mu \psi_i^L + \overline{\psi}_i^R \gamma^\mu \psi_i^R$$

where we have used $Q_i = T^3 + Y$ with T^3 giving 0 when acting on right-handed states.

6.2 Fermion masses

- To write down an electron mass, we can use the Higgs doublet; then the masses appear only after electroweak symmetry breaking. For example, the term $\mathcal{L}_{\text{Yuk}} = -y\overline{L}He_R + h.c.$ will generate a mass term $-m_e(\overline{e}_L e_R + \overline{e}_R e_L)$ with $m_e = \frac{y}{\sqrt{2}}v$ after H gets a vev. With this construction, the charged leptons and the down-type quarks (d, s, b) will get masses, and no additional $SU(2)$ breaking is required.

- To give mass to the remaining fermions, we can use $\overline{L}\sigma_2 H^\star$. This is $SU(2)$ invariant. We define $\widetilde{H} = i\sigma_2 H^\star$¡ which transforms in the fundamental representation of $SU(2)$ and has hypercharge $-\frac{1}{2}$. Then we can write $-y\overline{L}\widetilde{H}\nu_R$ as a term that gives a mass to the neutrino (or the up-type quarks).

- Focusing on the first quarks, including all three generations, indexed by i and j, we then have

$$\mathcal{L}_{\text{mass}} = -Y_{ij}^d \overline{Q}^i H d_R^j - Y_{ij}^u \overline{Q}^i \widetilde{H} u_R^j + h.c.$$

Note that each term is $SU(3) \times SU(2) \times U(1)$-invariant.

- After symmetry breaking, the quark mass terms become

$$\mathcal{L}_{\text{mass}} = -\frac{v}{\sqrt{2}} Y^d_{ij} \overline{d}^i_L d^j_R - \frac{v}{\sqrt{2}} Y^u_{ij} \overline{u}^i_L u^j_R + h.c. = -\frac{v}{\sqrt{2}} [\overline{d}_L Y_d d_R + \overline{u}_L Y_u u_R] + h.c.$$

  To diagonalize masses, we use that there exist two diagonal matrices M_d, M_u and two unitary matrices U_d, U_u for which $Y_d Y_d^\dagger = U_d M_d^2 U_d^\dagger, Y_u Y_u^\dagger = U_u M_u^2 U_u^\dagger$. The matrix $YY^\dagger$ is Hermitian; we can also generically write $Y_d = U_d M_d K_d^\dagger, Y_u = U_u M_u K_u^\dagger$ for other unitary matrices K_d, K_u. Thus the Yukawa couplings are

$$\mathcal{L}_{\text{mass}} = -\frac{v}{\sqrt{2}} [\overline{d}_L U_d M_d K_d^\dagger d_R + \overline{u}_L U_u M_u K_u^\dagger u_R] + h.c.$$

- Now we can freely change basis for the right-handed quarks by $d_R \to K_d d_R$, $u_R \to K_u u_R$ and the left-handed quarks by $u_L \to U_u u_L$, $d_L \to U_d d_L$, removing the U, K matrices from the Yukawa terms and leaving the diagonal mass matrices M_u, M_d. This is known as going to the *mass basis*; the terms are then just

$$\mathcal{L}_{\text{mass}} = -m^d_j \overline{d}^j_L d^j_R - m^u_j \overline{u}^j_L u^j_R + h.c.$$

  where m^d_j, m^u_j are the diagonal elements of $\frac{v}{\sqrt{2}} M_d$, $\frac{v}{\sqrt{2}} M_u$ respectively. Note that there is still a residual $U(1)^6$ global symmetry (e.g. $d^j_L \to e^{i\alpha_j} d^j_L$, with no sum on j, and so on).

- The gauge boson interactions do not mix families in the original *flavor basis* where the Lagrangian is

$$\begin{aligned}\mathcal{L}_{\text{flavor-basis}} = (\overline{u}_L \ \overline{d}_L)^i \left[i\partial\!\!\!/ + \gamma_\mu \begin{pmatrix} \frac{g'}{6} B_\mu + \frac{g}{2} W^3_\mu & \frac{g}{\sqrt{2}} W^+_\mu \\ \frac{g}{\sqrt{2}} W^-_\mu & \frac{g'}{6} B_\mu - \frac{g}{2} W^3_\mu \end{pmatrix} \right] \begin{pmatrix} u_L \\ d_L \end{pmatrix}^i \\ + \overline{u}^i_R \left(i\partial\!\!\!/ + g' \frac{2}{3} B\!\!\!/ \right) u^i_R + \overline{d}^i_R \left(i\partial\!\!\!/ - g' \frac{1}{3} B\!\!\!/ \right) d^i_R \\ - \frac{v}{\sqrt{2}} \left[\overline{d}^i_L (U_d M_d K_d^\dagger)_{ij} d^j_R + \overline{u}^i_L (U_u M_u K_u^\dagger)_{ij} u^j_R + h.c. \right]\end{aligned}$$

  where i, j are flavor indices; when we rotate $d^i_R \to K_d^{ij} d^j_R$ and $u^i_R \to K_u^{ij} u^j_R$, the matrices K_u, K_d drop out completely since the hypercharge interactions are generation diagonal; when we rotate $u_L \to U_u u_L, d_L \to U_d d_L$, the B_μ and W^3_μ couplings are unaffected as well, since these do not mix up- and down-type quarks. The only things that are sensitive to flavor rotations are the $W^\pm_\mu$ couplings. Thus we have

$$\begin{aligned}\mathcal{L}_{\text{mass-basis}} = \frac{e}{\sin\theta_w} Z_\mu J^Z_\mu + e A_\mu J^\mu_{\text{EM}} - m^d_j (\overline{d}^j_L d^j_R + \overline{d}^j_R d^j_L) - m^u_j (\overline{u}^j_L u^j_R + \overline{u}^j_R u^j_L) \\ + \frac{e}{\sqrt{2}\sin\theta_w} [W^+_\mu \overline{u}^i_L \gamma^\mu (V)^{ij} d^j_L + W^-_\mu \overline{d}^i_L \gamma^\mu (V^\dagger)^{ij} u^j_L]\end{aligned}$$

  where $V = U_u^\dagger U_d$.

- Thus, all of the interesting mixing effects are given by a single matrix, known as the *Cabibbo-Kobayashi-Maskawa (CKM) matrix*:

$$V \equiv U_u^\dagger U_d = \begin{pmatrix} V_{11} & V_{12} & V_{13} \\ V_{21} & V_{22} & V_{23} \\ V_{31} & V_{32} & V_{33} \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix},$$

  The CKM matrix is a complex unitary matrix and thus has nine real degrees of freedom.

- A nonzero phase in the CKM matrix implies CP violation.

7 Second quantization

7.1 Quantum harmonic oscillator

- Classically, for a nonrelativistic mass M in a simple harmonic oscillator whose position is given by $\phi(t)$, normal modes look like $\phi(t) = Ae^{-i\omega_0 t} + A^* e^{i\omega_0 t}$. We rewrite this as $\phi(t) = \frac{1}{\sqrt{2M\omega_0}}(ae^{-i\omega_0 t} + a^* e^{i\omega_0 t})$ for reasons that become clear when we quantize. The (conjugate) momentum is $p(t) = \sqrt{\frac{M}{2\omega_0}}(-i\omega_0 a e^{-i\omega_0 t} + i\omega_0 a^* e^{i\omega_0 t})$.

- In quantum mechanics, possible states are represented by rays. Call $|y\rangle$ the definite ϕ state $|\phi = y\rangle$. The set $\{|y\rangle\}$ for all possible y forms a basis for the Hilbert space, in the sense that the most general state $|\psi\rangle$ is expressible as a superposition of these with some coefficients $|\psi\rangle = \sum_y \Psi(y)\,|y\rangle = \int_{-\infty}^{\infty} dy \Psi(y)\,|y\rangle$.

- The coefficient $\Psi(y)$ has the interpretation as the probability amplitude; $\Psi(y) = \langle y\,|\psi\rangle$.

- Quantization corresponds to promoting c-number-valued physical observables to linear, Hermitian operators acting on the Hilbert space.

- The harmonic oscillator Hamiltonian $H = \frac{1}{2M}p^2 + \frac{1}{2}\omega_0^2\phi^2$. When we quantize, becomes the Hamiltonian operator $H = \frac{\widehat{p}^2}{2M} + \frac{1}{2}\omega_0^2\widehat{\phi}^2$.

- Define the *ladder operators*

$$\widehat{a} = \sqrt{\frac{M\omega_0}{2}}\left(\widehat{\phi} + \frac{i}{M\omega_0}\widehat{\rho}\right)$$
$$\widehat{a}^\dagger = \sqrt{\frac{M\omega_0}{2}}\left(\widehat{\phi} - \frac{i}{M\omega_0}\widehat{\rho}\right)$$

  These satisfy the commutation relations $[\widehat{a}, \widehat{a}^\dagger] = 1$. We can rewrite the Hamiltonian operator as $\widehat{H} = \omega_0\left(\widehat{a}^\dagger\widehat{a} + \frac{1}{2}\right)$.

- The time-independent Schrodinger equation $\widehat{H}\,|\psi\rangle = E\,|\psi\rangle$ gives the energy eigenvalues $E_n = \omega_0\left(n + \frac{1}{2}\right)$ for $n \in \mathbb{N}$. We denote the ground state $|0\rangle$ and higher eigenstates $|n\rangle$. We define the *renormalized Hamiltonian operator* $\widehat{H} = \omega_0\widehat{a}^\dagger\widehat{a}$ by subtracting the zero-point energy, so that the energy spectrum becomes $E_n = \omega_0 n$. From now on, by "Hamiltonian $\widehat{H}$" we refer to the renormalized Hamiltonian operator.

- We also have the commutation relations $[\widehat{a}, \widehat{H}] = \omega_0\widehat{a}$, $[\widehat{a}^\dagger, \widehat{H}] = -\omega_0\widehat{a}^\dagger$.

- $\widehat{a}^\dagger$ raises states, and $\widehat{a}$ lowers them; $\widehat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$ for $n \geq 0$, $\widehat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$ for $n > 0$, and $\widehat{a}\,|0\rangle = 0$ (the coefficients in front of the eigenstates come from normalizing).

- There is also the *number operator* $\widehat{N} = \widehat{a}^\dagger\widehat{a}$, which counts modes: $\widehat{N}\,|n\rangle = n\,|n\rangle$.

- In quantum field theory, it is useful to use the definite energy basis $|\psi\rangle = \sum_{n=0}^{\infty}\frac{c_n}{\sqrt{n}}(\widehat{a}^\dagger)^n\,|0\rangle$.

7.2 Time evolution and the Heisenberg picture

- While introductory quantum mechanics works in the Schrodinger picture (state vectors evolve in time, operators are mostly constant with respect to time), in quantum field theory we work in the *Heisenberg picture* (states kept constant, observables involve in time).

- In the Schrodinger picture, a state ket $|\psi(t)\rangle$ evolves as $|\psi(t)\rangle = \widehat{U}(t)\,|\psi(0)\rangle$, where $\widehat{U}(t) = \exp(-i\widehat{H}t)$.

- It is convenient to give the raising/lowering operators explicit time dependence by simply defining

$$\widehat{a}_S(t) = e^{i\omega_0 t}\widehat{a}_S(0) = e^{i\omega_0 t}\widehat{a}$$
$$\widehat{a}_S^\dagger(t) = e^{-i\omega_0 t}\widehat{a}_S^\dagger(0) = e^{-i\omega_0 t}\widehat{a}^\dagger$$

The (time-independent) position and momentum operators then take the form

$$\widehat{\phi}_S = \frac{1}{\sqrt{2M\omega_0}}(\widehat{a}_S(t)e^{-i\omega_0 t} + \widehat{a}_S^\dagger(t)e^{i\omega_0 t})$$
$$\widehat{p}_S = \sqrt{\frac{M}{2\omega_0}}(-i\omega_0\widehat{a}_S(t)e^{-i\omega_0 t} + i\omega_0\widehat{a}_S^\dagger(t)e^{i\omega_0 t})$$

Note that there is no time-dependence, since $\widehat{a}_S(t)e^{-i\omega_0 t} = \widehat{a}$, $\widehat{a}_S^\dagger(t)e^{i\omega_0 t} = \widehat{a}^\dagger$. Compare this with our classical expressions: the quantum operators corresponding to classical mode amplitudes a, a^* are the raising/lowering operators $\widehat{a}, \widehat{a}^\dagger$ for the normal mode.

- In the Heisenberg picture, the state ket is constant, and operators evolve as $\widehat{A}_H(t) = \widehat{U}^\dagger \cdot \widehat{A}_S(t) \cdot \widehat{U}$.

- Hence the Heisenberg picture field operators have the mode expansion

$$\widehat{\phi}_H = \frac{1}{\sqrt{2M\omega_0}}(\widehat{a}e^{-i\omega_0 t} + \widehat{a}^\dagger e^{i\omega_0 t})$$
$$\widehat{p}_H = \sqrt{\frac{M}{2\omega_0}}(-i\omega_0\widehat{a}e^{-i\omega_0 t} + i\omega_0\widehat{a}^\dagger e^{i\omega_0 t})$$

From now on, we work in the Heisenberg picture by default, leaving the labels "H" implicit.

7.3 Field expansion

- The most general solution of the equation of motion for a classical string of length L is some linear combination of normal modes of simple harmonic oscillators $\phi(t,x) = \sum_{n=-\infty}^{\infty} A_n e^{-i\omega_n t + ik_n x} + A_n^* e^{i\omega_n t - ik_n x}$ where $k_n = \frac{2\pi n}{L}$, $n \in \mathbb{Z}$, and $\omega_n = \sqrt{\frac{T}{\rho}k_n^2 + \omega_0^2}$.

- Define the quantum mode operators at any time t by

$$\widehat{\phi}(t,x) = \sum_{n=-\infty}^{\infty}\sqrt{\frac{1}{2\omega_n L}}[\widehat{a}_n(t)e^{-i\omega_n t + ik_n x} + \widehat{a}_n^\dagger(t)e^{i\omega_n t - ik_n x}]$$
$$\widehat{\pi}(t,x) = (-i\omega_n)\sum_{n=-\infty}^{\infty}\sqrt{\frac{1}{2\omega_n L}}[\widehat{a}_n(t)e^{-i\omega_n t + ik_n x} - \widehat{a}_n^\dagger(t)e^{i\omega_n t - ik_n x}]$$

using the Fourier identities $\delta_{mn} = \frac{1}{L}\int_0^L dx e^{i(k_m - k_n)x}$, $\delta(x - x') = \frac{1}{L}\sum_{n=-\infty}^{\infty} e^{ik_n(x-x')}$, we get an expression for the ladder operators in terms of the mode operators:

$$\widehat{a}_n(t) = e^{i\omega_n t}\sqrt{\frac{\omega_n}{2L}}\int_0^L dx e^{-ik_n x}\left[\widehat{\phi}(t,x) + \frac{i}{\omega_n}\widehat{\pi}(t,x)\right]$$
$$\widehat{a}_n^\dagger(t) = e^{-i\omega_n t}\sqrt{\frac{\omega_n}{2L}}\int_0^L dx e^{ik_n x}\left[\widehat{\phi}(t,x) - \frac{i}{\omega_n}\widehat{\pi}(t,x)\right]$$

For the free Hamiltonian, the implicit time-dependence of $\widehat{\phi}(t,x), \widehat{\pi}(t,x)$ in the Heisenberg picture means that the mode operators are actually time-independent.

- We have the commutation relations $[\widehat{a}_m, \widehat{a}_n^\dagger] = \delta_{mn}$.

- Each Noether charge of classical field theory becomes a quantum operator. Specifically, plugging in the mode expansion,

$$\widehat{H} = \int_0^L dx \widehat{T}^{00} - \delta = \sum_{n=-\infty}^{\infty} \omega_n \widehat{a}_n^\dagger \widehat{a}_n \widehat{P} \qquad = \int_0^L dx \widehat{T}^{01} = \sum_{n=-\infty}^{\infty} k_n \widehat{a}_n^\dagger \widehat{a}_n$$

  where δ is a renormalization constant.

- Taking our one-dimensional string, turning it into a box of length L, and taking the limit $L \to \infty$, we get the expression for a *quantum scalar field*:

$$\widehat{\phi}(t, \vec{x}) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} (\widehat{a}_{\vec{k}}(t) e^{-ik\cdot x} + \widehat{a}_{\vec{k}}^\dagger(t) e^{ik\cdot x}$$

  with $\omega_{\vec{k}} = \sqrt{|\vec{k}|^2 + m^2}$.

- Discrete sums over normal modes become continous integrals over wavenumber.

- The discrete commutation relations for mode operators become continuous commutation relations: $[\widehat{a}_{\vec{k}}(t), \widehat{a}_{\vec{k}'}^\dagger(t)] = (2\pi)^3 \delta^3(\vec{k} - \vec{k}')$.

- The *Hamiltonian for the free theory* is thus

$$\widehat{H} = \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} \widehat{a}_{\vec{k}}^\dagger \widehat{a}_{\vec{k}}$$

- This is known as *second quantization*. First quantization refers to the discrete modes, for example, of a particle in a box. Second quantization refers to integer numbers of excitations of each of these modes. This is somewhat misleading: the fact that there are discrete modes is a classical phenomenon. The two steps really are (1) interpret these modes as having energy $E = \omega$ and (2) quantize each mode as a harmonic oscillator. In that sense, we are only quantizing once.

- In second quantization, we have many quantum mechanical systems–one for each $\widehat{k}$–all at the same time. We interpret the nth excitation of the $\widehat{k}$ harmonic oscillator as having n *particles*.

- In second quantization, the Hilbert space is promoted to a *Fock space* which is defined as a direct sum $\mathcal{F} = \bigoplus_n \mathcal{H}_n$ of Hilbert spaces $\mathcal{H}_n$ of physical n-particle states. If there is one particle type, states in $\mathcal{H}_n$ are linear combinations of states $\{|k_1^\mu, \cdots, k_n^\mu\rangle\}$ of all possible momenta satisfying $k_i^2 = m^2$ with $k_i^0 > 0$. If there are many different particle types, the Fock space is the direct sum of the Hilbert spaces associated with each particle. The Fock space is time-independent by time-translation invariance, and frame-invariant, by Lorentz invariance. note that the Fock space is not a sum over Hilbert spaces defined with arbitrary 4-vectors, since the energy for a physical state is determined by its 3-momentum $\vec{k}_i$ and its mass m_i as $k_i^0 = \sqrt{\vec{k}_i^2 + m_i^2}$.

- (Multi) particle states are produced by acting with raising mode operators on a vacuum: $|0\rangle \to \sqrt{2\omega_{\vec{k}}} \widehat{a}_{\vec{k}}^\dagger |0\rangle$.

8 Spin and statistics

9 Scattering amplitudes

9.1 Feynman rules

10 Electromagnetic duality

10.1 The Dirac monopole

- Write the Maxwell's equations

$$\vec{\partial} \cdot \vec{E} = 0, \quad \vec{\partial} \cdot \vec{0}, \quad \vec{\partial} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \vec{\partial} \times \vec{B} = \frac{\partial \vec{E}}{\partial t}$$

These are highly symmetric; they are invariant under Lorentz transformations (in fact, conformal) and *electromagnetic duality* $(\vec{E}, \vec{B}) \mapsto (\vec{B}, -\vec{E})$.

- Take the definition of $F_{\mu\nu}$ in terms of the field components (i.e. $F^{0i} = -F^{0i} = -E^i$, $F^{ij} = -\epsilon_{ijk} B^k$). In terms of $F_{\mu\nu}$, we can write Maxwell's equations

$$\partial_\nu F^{\mu\nu} = 0, \quad \partial_\nu^\star F^{\mu\nu} = 0$$

where

$$^\star F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\lambda\rho} F_{\lambda\rho}$$

The duality transformation is then simply $F^{\mu\nu} \mapsto^\star F^{\mu\nu}$, $^\star F^{\mu\nu} \mapsto -F^{\mu\nu}$. In Minkowski space, $\partial_\nu^\star F^{\mu\nu} = 0$ implies that $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\partial_\mu F^{\mu\nu} = 0$ implies that $^\star F_{\mu\nu} = \partial_\nu \tilde{A}_\nu - \partial_\nu \tilde{A}_\mu$ for some dual electromagnetic field $\tilde{A}_\mu$. Notice that the duality transformation relating A_μ and $\tilde{A}_\mu$ is nonlocal.

- In the presence of sources, duality is preserved provided that we include both electric and magnetic sources,

$$\partial_\nu F^{\mu\nu} = j^\nu, \quad \partial_\nu^\star F^{\mu\nu} = k^\mu$$

and that we supplement that duality transformations by a similar transformation of the sources $j^\mu \mapsto k^\mu$, $k^\mu \mapsto -j^\mu$.

- A charged point particle in the presence of an electromagnetic field behaves according to the Lorentz force law. If the particle is also magnetically charged, the Lorentz force is given by

$$m \frac{d^2 x^\mu}{d\tau^2} = (q F^{\mu\nu} + g^\star F^{\mu\nu}) \frac{dx_\mu}{d\tau}$$

where q, g are electric, magnetic charges, respectively; this formula is invariant under duality provided we also interchange $(q, g) \mapsto (g, -q)$.

- In the presence of magnetic sources, $\partial_\nu^\star F^{\mu\nu} \neq 0$ when there is no A_μ. Nevertheless, if the magnetic sources are localized in space at a given moment in time, one may define A_μ in those regions where $k^\mu = 0$; the topology of such regions is generically nontrivial and therefore a nonsingular A_μ need not exist throughout. Instead one solves A_μ locally, any two solutions being related, in their common domain of definition, by a gauge transformation.

- Place a *magnetic monopole*, i.e. a point-like source of a magnetic field, at the origin in $\mathbb{R}^3$. Then $\vec{B}(\vec{r}) = \frac{g}{4\pi} \frac{\vec{r}}{r^3}$, so that the flux through the unit sphere Σ is $g = \int_\Sigma \vec{B} \cdot d\vec{S}$. For $x \in \mathbb{R}^3 \backslash \{0\}$, $\vec{\partial} \times \vec{B} = 0$, so we can try to solve for some $\vec{A}$ with $\vec{B} = \vec{\partial} \times \vec{A}$. It happens to be that any $\vec{A}$ must be singular on some string-like region, called the *Dirac string*. For example, we could write $\vec{A}_+(\vec{r}) = \frac{g}{4\pi r} \frac{1 - \cos\theta}{\sin\theta} \hat{e}_\phi$, but it doesn't work for the negative z-axis $\theta = \pi$. We can define $\vec{A}_-$ by reversing the sign of the expression and the sign in the numerator, but then it works everywhere but on the positive z-axis $\theta = 0$.

- Since $\vec{\partial} \times (\vec{A}_+ - \vec{A}_-) = 0$ on $\mathbb{R}^3 \backslash \{\phi = n\pi/2\}$, we'd expect some χ s.t. $\vec{\partial}\chi = \vec{A}_+ - \vec{A}_-$, but this region is not simply connected, and χ need only be defined locally.

- Suppose that we are quantizing a particle of mass m and charge q in the field of a magnetic monopole. The Schrodinger equation satisfied by the wavefunction is $-\frac{\hbar}{2m}\nabla^2\psi = i\hbar\frac{\partial\psi}{\partial t}$, where $\vec{\nabla} = \vec{\partial} + ie\vec{A}$, for $e = q/\hbar$. The Schrodinger equation is invariant under gauge transformations $\vec{A} \mapsto \vec{A} + \vec{\partial}\chi$, $\psi \mapsto \exp(-ie\chi)\psi$. Gauge invariance guarantees that solutions of the Schrodinger equation obtained locally with a particular $\vec{A}$ will patch up nicely, provided that the wavefunction be single-valued. This is equivalent to the *Dirac quantization condition*:
$$eg = 2\pi n \quad \text{for some } n \in \mathbb{Z}$$

- The orbital angular momentum $\vec{L} = \vec{r} \times m\dot{\vec{r}}$ of a particle of mass m, charge q in the presence of a magnetic monopole is not conserved. By Lorentz force law, $\frac{d\vec{L}}{dt} = \frac{d}{dt}\left(\frac{qg}{4\pi}\frac{\vec{r}}{r}\right)$, so the conserved quantity is instead $\vec{J} \equiv \vec{L} - \frac{qg}{4\pi}\frac{\vec{r}}{r}$. The correction term is nothing but the angular momentum due to the electromagnetic field, where $\vec{J}_{\text{em}} = \int_{\mathbb{R}^3} d^3r\vec{r} \times (\vec{E} \times \vec{B})$; if we assume that the electromagnetic angular momentum is separately quantized, so that $|\vec{J}_{\text{em}}| = \frac{1}{2}n$ for some $n \in \mathbb{Z}$, we recover the Dirac quantization.

- A *dyon* is a particle which possesses both electric and magnetic charge. Consider two dyons of charges (e, g) and (e', g'); imposing the quantization of angular momentum of the resulting electromagnetic field yields the *Zwanziger-Schwinger quantization condition*
$$eg' - e'g = 2\pi n \quad \text{for some } n \in \mathbb{Z}$$

10.2 The 't Hooft-Polyakov monopole

- The Georgi-Glashow model was an early proposal to describe electroweak interactions. We will consider only the bosonic part of the model which consists of an $SO(3)$ Yang-Mills field theory coupled to a Higgs field in the adjoint representation. The Lagrangian density is given by
$$\mathcal{L} = -\frac{1}{4}\vec{G}^{\mu\nu}\vec{G}_{\mu\nu} + \frac{1}{2}D^\mu\vec{\phi}D_\mu\vec{\phi} - V(\phi)$$
where
  - The gauge field strength
$$\vec{G}_{\mu\nu} = \partial_\mu\vec{W}_\nu - \partial_\nu\vec{W}_\mu - e\vec{W}_\mu \times \vec{W}_\nu$$
where $\vec{W}_\mu$ are gauge potentials taking values in the Lie algebra of $SO(3)$, which we identify with $\mathbb{R}^3$ with the cross product for Lie bracket.
  - The Higgs field $\vec{\phi}$ is a vector in the 3D adjoint representation of $SO(3)$, with components $\phi_a = (\phi_1, \phi_2, \phi_3)$ which is minimally coupled to the gauge field via the gauge-covariant derivative
$$D_\mu\vec{\phi} = \partial_\mu\vec{\phi} - e\vec{W}_\mu \times \vec{\phi}$$
  - The Higgs potential $V(\phi)$ is given by
$$V(\phi) = \frac{1}{4}\lambda(\phi^2 - a^2)^2$$
where $\phi^2 = \vec{\phi} \cdot \vec{\phi}$ and λ is assumed non-negative.

- $\mathcal{L}$ is invariant under the $SO(3)$ gauge transformations
$$\vec{\phi} \mapsto \vec{\phi}' = g(x)\vec{\phi}$$
$$\vec{W}_\mu \mapsto \vec{W}'_\mu = g(x)\vec{W}_\mu g(x)^{-1} + \frac{1}{e}\partial_\mu g(x)g(x)^{-1}$$
where $g(x) \in SO(3)$.

- The classical dynamics of the fields $\vec{W}_\mu$, $\vec{\phi}$ are determined from the equations of motion

$$D_\nu \vec{G}^{\mu\nu} = -e\vec{\phi} \times D^\mu \vec{\phi}$$
$$D^\mu D_\mu \vec{\phi} = -\lambda(\phi^2 - a^2)\vec{\phi}$$

  and by the Bianchi identity $D_\mu {}^\star\vec{G}^{\mu\nu} = 0$ where ${}^\star\vec{G}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\lambda\rho}\vec{G}_{\lambda\rho}$.

- The canonically conjugate momenta to the gauge field $\vec{W}_\mu$ and the Higgs $\vec{\phi}$ are given by $\vec{E}^i = -\vec{G}^{0i}$, $\vec{\Pi} = D_0\vec{\phi}$. Defining $\vec{B}_i$ by $\vec{G}_{ij} = -\epsilon_{ijk}\vec{B}^k = \epsilon_{ijk}\vec{B}_k$, we can write the energy density as

$$\mathcal{H} = \frac{1}{2}\vec{E}_i \cdot \vec{E}_i + \frac{1}{2}\vec{\Pi} \cdot \vec{\Pi} + \frac{1}{2}\vec{B}_i \cdot \vec{B}_i + \frac{1}{2}D_i\vec{\phi} \cdot D_i\vec{\phi} + V(\phi)$$

  which is positive-semidefinite and gauge-invariant.

- Define a *vacuum configuration* to be one for which the energy density vanishes, meaning that $\vec{G}_{\mu\nu} = D^\mu\vec{\phi} = V(\phi) = 0$. For example, $\vec{\phi} = a\hat{e}_3$, $\vec{W}_\mu = 0$, where $(\hat{e}_a)$ is an orthonormal basis for the three-dimensional representation space where the Higgs field takes valued. We also define the *Higgs vacuum* as those coonfigurations of the Higgs field which satisfy the latter two equations above. Notice that in the Higgs vacuum, the Higgs field obeys $\phi^2 = a^2$. Any such vacuum configuration is not invariant under the whole $SO(3)$, but only under an $SO(2) \cong U(1)$ subgroup, therefore this model exhibits spontaneous symmetry breaking.

- Let $\vec{\phi} = \vec{a} + \vec{\varphi}$ where $\vec{a}$ is a constant vector obeying $\vec{a} \cdot \vec{a} = a^2$. The model then consists of a massless vector boson $A_\mu = \frac{1}{a}\vec{a} \cdot \vec{W}_\mu$, which we identify with the photon, a massive scalar field $\varphi = \frac{1}{a}\vec{a} \cdot \vec{\phi}$, and two massive vector bosons $W^\pm_\mu$. The masses of φ and $W^\pm_\mu$ are read off from the quadratic terms of the Lagrangian density to be $M_H = a\sqrt{2\lambda}$ and $M_W = ae$ respectively, and the charge of $W^\pm_\mu$ read off from the coupling to the photon is $\pm e$.

- The energy of a given field configuration is the spatial integral $E = \int d^3x\mathcal{H}$. Finite-energy solutions to the equations of motion, then, must approach a vacuum configuration asymptotically. In particular, the Higgs field approaches the Higgs vacuum at spatial infinity.

- Let us think of the Higgs potential V as a function $V : \mathbb{R}^3 \to \mathbb{R}$, and define $M_0 = \{\vec{x} \in \mathbb{R}^3 : V(\vec{x}) = 0\} \subset \mathbb{R}^3$. In the model at hand, M_0 is the sphere of radius a, hence in any finite-energy configuration the Higgs field defines a function from the sphere at spatial infinity to M_0, $\vec{\phi}_\infty(\hat{r}) \equiv \lim_{r\to\infty}\vec{\phi}(\vec{r}) \in M_0$. We can assume that $\vec{\phi}_\infty$ is continuous.

- The *topological number* of a finite-energy configuration is defined to be the degree[15] of the map $\vec{\phi}_\infty$. The zero energy vacuum configuration $\vec{W}_\mu = 0$, $\vec{\phi} = a\hat{e}_3$ has zero degree since $\vec{\phi}_\infty$ is constant. The topological number of a field configuration is invariant under any continuous deformation. In particular, it is invariant under time evolution and under gauge transformations, since the gauge group is connected. hence if we set up a finite-energy field configuration at some moment in time whose topological number is nonzero, it will never dissipate (i.e. will never evolve in time towards a trivial solution).

- A solution is defined to be *static* if it is time-independent and in addition $\vec{W}_0 = 0$. (Note that this is not merely a choice of gauge, since the gauge transformation to $\vec{W}_0 = 0$ is time-dependent, which is not allowed since we are looking for time-independent solutions). For static configurations, $\vec{E}$ and $\vec{\Pi}$ both vanish, hence the energy agrees up to a sign with the Lagrangian. Therefore a field configuration will be a solution iff it extremizes the energy.

[15] For a continuous map $f : S^n \to S^n$, there is an induced pushforward homomorphism $f_* : H_n(S^n) \to H_n(S^n)$; since $H_n(S^n) \cong \mathbb{Z}$, f_* must be of the form $f_* : x \mapsto \alpha x$ for some fixed $\alpha \in \mathbb{Z}$, called the degree of f.

- The *'t Hooft-Polyakov ansatz* for the monopole is given by

$$\begin{aligned}\vec{\phi}(\vec{r}) &= \frac{\vec{r}}{er^e}H(aer)\\ W_a^i &= -\epsilon_{aij}\frac{r^j}{er^2}(1-K(aer))\\ W_a^0 &= 0\end{aligned}$$

  for some arbitrary functions H and K. Plugging the ansatz into the energy density, we find that the integral for energy exists provided that $K \to 0$ and $H/\xi \to 1$ sufficiently fast as $\xi \to \infty$. Specifically, $K - 1 \leq O(\xi)$ and $H \leq O(\xi)$ as $\xi \to 0$. With these boundary conditions, $\vec{\phi}_\infty(\widehat{r}) = a\widehat{r}$, which is homotopic to the identity map and hence has degree 1. If such a solution exists, it is therefore stable and non-dissipative. We get the equations of motion for H and K:

$$\begin{aligned}\xi\frac{d^2K}{d\xi^2} &= KH^2 + K(K^2-1)\\ \xi\frac{d^2H}{d\xi^2} &= 2K^2H + \frac{\lambda}{e^2}H(H^2-\xi^2)\end{aligned}$$

  The boundary conditions are then $K \sim \exp(-\xi) = \exp(-M_W r)$ and $H - \xi \sim \exp(-M_H r)$.

- The electromagnetic vector potential $A_\mu = \frac{1}{a}\vec{\phi}\cdot\vec{W}_\mu$ corresponds to the $U(1) \subset SO(3)$ defined as the stabilizer of $\vec{\phi}$. The electromagnetic field can therefore be identified with $F_{\mu\nu} = \frac{1}{a}\vec{\phi}\cdot\vec{G}_{\mu\nu}$. Because the 't Hooft-Polyakov ansatz corresponds to a static solution, there is no electric field ($F_{0i} = 0$), but there is a magnetic field; in fact, the asymptomatic form of F_{ij} gives us that the asymptomatic field is that of the magnetic monopole $\vec{B} = -\frac{1}{e}\frac{\vec{r}}{r^3}$.

- The magnetic charge of a 't Hooft-Polyakov monopole is (up to a sign) twice the minimum magnetic charge consistent with the electric charge e and the Dirac quantization condition, i.e. twice the *Dirac charge* corresponding to e. This follows from the way that the electromagnetic $U(1)$ is embedded in $SO(3)$. We see that the 't Hooft-Polyakov solution describes an object of finite size which from far away cannot be distinguished from a Dirac monopole of charge $-4\pi/e$. In contrast with the Dirac monopole, the 't Hooft-Polyakov monopole is everywhere smooth, due to massive fields which become relevant as we approach the core of the monopole. As such, the magnetic charge is purely topological and due completely to the behavior of the Higgs field far away from the core.

- By the exponential decay of the massive fields away from the core of the monopole, the Higgs field approaches the Higgs vacuum, i.e. it satisfies $D_\mu\vec{\phi} = 0$, $\vec{\phi}\cdot\vec{\phi} = a^2$ up to terms of order $O(\exp(-r/R)$ where R is the effective size of the monopole, governed by the mass of the heavy particles. It is therefore reasonable to assume that any finite-energy solution of the Yang-Mills-Higgs system satisfies the Higgs vacuum conditions except in a finite number of well-separated compact localized regions in space which we call *monopoles*.

- In the Higgs vacuum, $\vec{\phi}\times\vec{W}_\mu = -\frac{1}{e}\partial_\mu\vec{\phi}$, whence $\vec{W}_\mu$ is fully determined except for the component in the $\vec{\phi}$ direction, which we denote by A_μ. We find that $\vec{W}_\mu = \frac{1}{a^2e}\vec{\phi}\times\partial_\mu\vec{\phi} + \frac{1}{a}\vec{\phi}A_\mu$. The field strength in the Higgs vacuum points in the $\vec{\phi}$ direction and is given by $\vec{G}_{\mu\nu} = \frac{1}{a}\vec{\phi}F_{\mu\nu}$ where $F_{\mu\nu} = \frac{1}{a^3e}\vec{\phi}\cdot(\partial_\mu\vec{\phi}\times\partial_\nu\vec{\phi}) + \partial_\mu A_\nu - \partial_\nu A_\mu$. This $F_{\mu\nu}$ satisfies Maxwell's equations.

- Let Σ be a surface in the Higgs vacuum enclosing some monopoles. The magnetic flux through Σ determines the magnetic charge. Notice that A_μ doesn't contribute, and that we get

$$g_\Sigma = \int_\Sigma \vec{B}\cdot d\vec{S} = -\frac{1}{2ea^2}\int_\Sigma \epsilon_{ijk}\vec{\phi}\cdot(\partial_j\vec{\phi}\times\partial_k\vec{\phi})dS_i$$

  Notice that only the components of $\partial_i\vec{\phi}$ tangential to Σ contribute to the integral and that magnetic charge therefore only depends on the behavior of $\vec{\phi}$ on Σ; furthermore, it only depends on the homotopy class of $\vec{\phi}$ as map $\Sigma \to M_0$; in other words, the above integral is invariant under deformations $\delta\vec{\phi}$ of $\vec{\phi}$ which preserve the Higgs vacuum. Examples of homotopies are time-evolution of $\vec{\phi}$, continuous gauge transformations on $\vec{\phi}$, and continuous changes of Σ within the Higgs vacuum.

- Write $g_\Sigma = -\frac{4\pi}{e}$ where $N_\Sigma = \frac{1}{8\pi a^3}\int_\Sigma dS_i \epsilon_{ijk}\vec{\phi}\times(\partial_j\vec{\phi}\times\partial_k\vec{\phi})$. This is the integral of the Jacobian of the map $\vec{\phi}:\Sigma\to M_0$, which is the classical definition of the degree of the map. Thus we recover the Dirac quantization condition $eg_\Sigma = -4\pi N_\Sigma$, with the same caveat as before about the fact that the minimum magnetic charge is twice the Dirac charge.

10.3 BPS-monopoles

- The mass of the source of a Dirac monopole is a free parameter. For the 't Hooft-Polyakov monopole there is no source, and the mass of the monopole is an intrinsic property of the Yang-Mills-Higgs system and as such it should be calculable. In the center of mass frame, all the energy of the monopole is concentrated in its mass

$$\begin{aligned} M &= \int_{\mathbb{R}^3}\left(\frac{1}{2}\vec{E}_i\cdot\vec{E}_i + \frac{1}{2}\vec{\Pi}\cdot\vec{\Pi} + \frac{1}{2}D_i\vec{\phi}\cdot D_i\vec{\phi} + V(\phi)\right) \\ &\geq \frac{1}{2}\int_{\mathbb{R}^3}\left(\vec{E}_i\cdot\vec{E}_i + \vec{B}_i\cdot\vec{B}_i + D_i\vec{\phi}\cdot D_i\vec{\phi}\right) \end{aligned}$$

where we have dropped some non-negative terms. Using some vector calculus, we obtain the lower bound for the mass is given by the *Bogomol'nyi bound*:

$$M \geq a\sqrt{q^2+g^2}$$

- States that saturate this bound are called *BPS states.*

- For static solutions, for saturation we require $V(\phi)$ should vanish and that the *Bogomol'nyi equation* should hold:

$$\vec{B}_i = \pm D_i\vec{\phi}$$

Now the only way to satisfy $V(\phi)=0$ and yet obtain a solution with a nonzero magnetic charge is for λ to vanish. We understand the condition $\lambda=0$ as a limiting value, and retain the boundary condition that at spatial infinity $\vec{\phi}$ satisfies $\vec{\phi}\cdot\vec{\phi}=a^2$. This is known as the *Prasad-Sommerfield limit.*

- It is not hard to find an explicit solution to the Bogomol'nyi equation in the 't Hooft-Polyakov ansatz; it yields $\xi\frac{dK}{d\xi} = -KH$, $\xi\frac{dH}{d\xi} = H+1-K^2$.

- One consequence of the Bogomol'nyi equation is that both the photon (through $\vec{B}_i$) and the Higgs (via $D_i\vec{\phi}$) contribute equally to the mass density. The long-range force exerted by the Higgs is always attractive and for static monopoles, it is equal in magnitude to the $1/r^2$ magnetic force. Therefore the forces add for oppositely charged monopoles, yet they cancel for equally charged monopoles. This is as it should be if static multi-monopole solutions saturating the Bogomol'nyi bound are to exist.

- The mass density at the origin for a BPS-monopole is finite.

10.4 Montonen-Olive duality and the Witten effect

- At $\lambda=0$, the bosonic spectrum of the Georgi-Glashow model (including BPS-monopoles) is given by:

  where $q = e\hbar$.

- Clearly, all particles satisfy the Bogomol'nyi bound, and the spectrum is invariant under electromagnetic $\mathbb{Z}_2$ duality $(q,g)\mapsto(g,-q)$ provided that we also interchange the BPS-monopoles and the massive vector bosons.

Particle	Mass	Electric Charge	Magnetic Charge	Spin/Helicity
Photon	0	0	0	± 1
Higgs	0	0	0	0
$W^{\pm}$ boson	aq	$\pm q$	0	1
$M^{\pm}$ monopole	aq	0	$\pm g$	0

- *Montonen-Olive duality* stems from a conjecture that there shuld be dual ("magnetic") description of this gauge theory where the elementary gauge particles are the BPS-monopoles and where the massive vector bosons appear as electric monopoles. Because of the Dirac quantization condition, if the coupling constant e of the original theory is small, the coupling constant g of the magnetic theory must be large, and vice versa. Hence, strong coupling behavior of a gauge theory can be determined by the weak coupling behavior of its dual theory. The conjecture was proven for $N = 4$ supersymmetric Yang-Mills theory.

- For the dyonic spectrum consistent with the quantization condition in a CP non-violating theory, one finds that the electric charge q of a dyon with minimal magnetic charge g could take one of two sets of mutually exclusive values: either $q = ne$ or $q = \left(n + \frac{1}{2}\right) e$, where $n \in \mathbb{Z}$. We see that it is the former case which holds in the following.

- Let N denote the operator which generated gauge transformations about $\vec{\phi}$:

$$\delta\vec{v} = \frac{1}{a}\vec{\phi} \times \vec{v}, \quad \delta\vec{W}_\mu = -\frac{1}{ea}D_\mu\vec{\phi}$$

  where $\vec{v}$ is any *isovector* (i.e. the vector transformation of a particle under the $SU(2)$ group of isospin). Consider the operator $\exp(2\pi i N)$. In the background of a finite-energy solution, the Higgs field is in the Higgs vacuum at spatial infinity, whence $\exp(2\pi i N)$ generates the identity transformation. On isovectors it generates a rotation about $\vec{\phi}$ of magnitude $2\pi|\vec{\phi}|/a = 2\pi$, and on the gauge fields we notice that $D_\mu\vec{\phi} = 0$ in the Higgs vacuum. Since $\exp(2\pi i N) = 1$, the eigenvalues of N are integral. To see what this means, we compute N.

- N is the charge of the Noether current associated with the above transformations. We find that $N = \frac{1}{ae}\int_{\mathbb{R}^3} \vec{E}_i \cdot D_i\vec{\phi} = \frac{q}{e}$. The quantization of N then implies that $q = ne$ for some $n \in \mathbb{Z}$.

- Introduce a θ-term $\mathcal{L}_\theta = -\frac{e^2\theta}{32\pi^2}{}^\star\vec{G}^{\mu\nu} \cdot \vec{G}_{\mu\nu}$. This term is locally a total derivative and hence does not contribute to the equations of motion. its integral in a given configuration is an integral multiple (called the *instanton number*) of the parameter θ. θ is therefor an angular variable and parametrizes inequivalent vacua. The Noether charge N becomes $N = \frac{q}{e} + \frac{e\theta}{8\pi^2}g$.

- For the 't Hooft-Polyakov monopole, $eg = -4\pi$, hence the integrality of N means that $q = ne + \frac{e\theta}{2\pi}$ for some $n \in \mathbb{Z}$. This result is called the *Witten effect*.

10.5 $SL(2, \mathbb{Z})$

- The action defined by $\mathcal{L} + \mathcal{L}_\theta$ depends on e, θ, λ, a. Introduce the complex linear combination $\vec{\mathcal{G}}_{\mu\nu} = \vec{G}_{\mu\nu} + i{}^\star\vec{G}_{\mu\nu}$. The dependence on the first two parameters can be unified into a complex paramater $\tau = \frac{\theta}{2\pi} + i\frac{4\pi}{e^2}$, in which case, after rescaling the gauge fields $\vec{W}_\mu \mapsto e\vec{W}_\mu$,

$$\begin{aligned}\mathcal{L} + \mathcal{L}_\theta &= -\frac{1}{4e^2}\vec{G}_{\mu\nu} \cdot \vec{G}^{\mu\nu} + \frac{\theta}{32\pi^2}\vec{G}_{\mu\nu} \cdot {}^\star\vec{G}^{\mu\nu} + \frac{1}{2}D^\mu\vec{\phi} \cdot D_\mu\vec{\phi} - V(\phi) \\ &= -\frac{1}{32\pi}\mathrm{Im}\left(\tau\vec{\mathcal{G}}_{\mu\nu} \cdot \vec{\mathcal{G}}^{\mu\nu}\right) + \frac{1}{2}D^\mu\vec{\phi} \cdot D_\mu\vec{\phi} - V(\phi)\end{aligned}$$

  Because θ is an angular variable, it is only defined up to 2π, so the physics is invariant under $\tau \mapsto \tau + 1$. At $\theta = 0$, Montonen-Olive tells us that $e \mapsto g = -4\pi/e$ is a symmetry. But this duality transformation is just $\tau \mapsto -1/\tau$. We are therefore tempted to strengthen the conjecture to say that for arbitrary θ, the physics should depend on τ only modulo

$$T : \tau \mapsto \tau + 1$$
$$S : \tau \mapsto -\frac{1}{\tau}$$

- $SL(2, \mathbb{Z})$ acts naturally on the complex plane:

  $$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}$$

  This action preserves the upper half-plane, so that if $\operatorname{Im} \tau > 0$, so will its transform under $SL(2, \mathbb{Z})$. The matrices $1, -1$ act trivially (and are the only two matrices that do), thus the action is not faithful, but it becomes faithful once we identify every matrix $M \in SL(2, \mathbb{Z})$ with $-M$; the resulting group is $PSL(2, \mathbb{Z}) \cong SL(2, \mathbb{Z})/\{\pm 1\}$.

- S and T as defined above generate $PSL(2, \mathbb{Z})$.

- The mass formula for BPS-states is invariant under $SL(2, \mathbb{Z})$.

- An improved Montonen-Olive conjecture states that physics is $SL(2, \mathbb{Z})$-invariant. This means that the theories defined by two values of τ related by the action of $SL(2, \mathbb{Z})$ are physically equivalent, provided that we are willing to relabel magnetic and electric charges by that same $SL(2, \mathbb{Z})$ transformation.

- There is a *fundamental domain*

  $$D = \left\{ \tau \in \mathbb{C} : \operatorname{Im} \tau > 0, |\operatorname{Re} \tau| \leq \frac{1}{2}, |\tau| \geq 1 \right\}$$

  which has the property that its orbit under $PSL(2, \mathbb{Z})$ span the whole upper half-plane and that no two points in its interior ($\left\{ \tau \in D : |\operatorname{Re} \tau| < \frac{1}{2}, |\tau| > 1 \right\}$) are related by the action of $PSL(2, \mathbb{Z})$.

- If we

Preface to the appendix

- Appendices A and B are based on Griffiths' treatment.
- Appendix C is based on Setter's notes, which follows Hall's treatment.
- Appendix D is based on an aside in Weinberg's textbook.
- Appendix E is based on deWitt's appendix.
- Appendix F is based on Budinich & Trautman's treatment, with notations borrowed from Bourbaki.

Appendix A: Elementary particles

What follows in the next two sections is largely based on Griffiths' text, and often taken verbatim. This gives the motivation for quantum field theory as the theory that governs/explains the forthcoming rules.

A.1 1900-1964

- *Electrons*, *protons*, and *neutrons* form atoms.
- The *photon* is the *quantum* of the *electromagnetic field*; electromagnetic interactions are mediated by an exchange of particles.
- The *strong force* holds the atomic nucleus together, preventing the protons from repelling each other. Yukawa predicted that the quantum of this field would be 300 times the mass of an electron, or a sixth of that of a proton.
- Yukawa's particle came to be known as the *meson* (middle-weight). The electron is called a *lepton* (light-weight) and the proton and neutron *baryons* (heavy-weight).
- Yukawa's particle was observed in cosmic rays and is called the π (*pion*).
- In searching for this particle, a lepton, the μ (*muon*) was also discovered.
- For every particle, there must exist a corresponding *antiparticle*, with the same mass but the opposite physical charge (such as electric charge). The standard notation is an overbar; for example, the proton p has the *antiproton* $\overline{p}$, and the neutron n has the *antineutron* $\overline{n}$. In some cases it is customary to simply specify the charge; the electron e (or e^-) has the *positron* e^+, and the μ (or μ^-) has the *antimuon* μ^+.
- Some neutral particles are their own antiparticles, for example the photon ($\overline{\gamma} \equiv \gamma$).

There is a general principle in particle physics called *crossing symmetry.* Suppose a reaction of the form $A + B \to C + D$ is known to occur; then, any of these particles can be 'crossed' over to the other side of the equation, provided it is turned into its antiparticle, and the resulting interaction will also be allowed. For example, the following are allowed:

$$
\begin{aligned}
A &\to \overline{B} + C + D \\
B + \overline{C} &\to \overline{B} + D \\
\overline{C} + \overline{D} &\to \overline{A} + \overline{B}
\end{aligned}
$$

This tells us, for instance, that Compton scattering $\gamma + e^- \to \gamma + e^-$ is really the same process as pair annihilation $e^- + e^+ \to \gamma + \gamma$, although in the laboratory they are completely different phenomena. Note that the crossed reaction is dynamically permissible, but it may or may not be kinematically allowed.

- In explaining varying emitted electron energy in beta decay, a new, light, electrically neutral particle, called the *neutrino* (meaning little one) and denoted ν, was discovered. Neutrinos are leptons. The fundamental beta decay process is then $n \to p^+ + e^- + \overline{\nu}$ The decay processes

  $$\pi \to \mu + \nu$$

  and $\mu \to e + 2\nu$ have also been observed.
- A rule of thumb in particle physics: whatever is not expressly forbidden is mandatory. Therefore, we must find reasons why some reactions are inadmissible.
- *Law of conservation of lepton number*: Assign a lepton number $L = +1$ to the electron, muon, and neutrino, $L = -1$ to the positron, positive muon, and antineutrino, and $L = 0$ to all other particles. In any physical process, the sum of lepton numbers before must be equal to the sum of lepton numbers after.

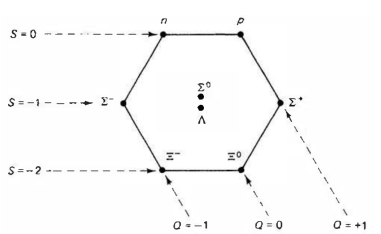
- The absence of $\mu \to e + \gamma$ suggests a law of conservation of mu-ness, but we observe $\mu \to e + \nu + \overline{\nu}$; this is because there are actually two different kinds of neutrino, the ν_e and the ν_μ, associated with the electron and muon respectively. If we assign a *muon number* $L_\mu = +1$ to μ^- and ν_μ and $L_\mu = -1$ to μ^+ and $\overline{\nu}_\mu$ (and 0 to everything else) and an *electron number* $L_e = +1$ to e^- and ν_e and $L_e = -1$ to e^+ and $\overline{\nu}_e$, then we can introduce the law of conservation of muon number and the law of conservation of electron number and everything works out.
- Under the refinements above, neutron beta decay becomes $n \to p^+ + e^- + \overline{\nu}_e$ and pion and muon decays take the form

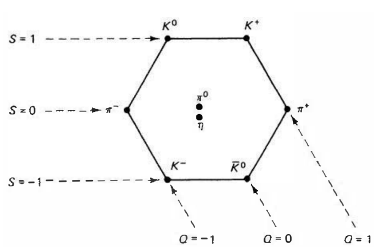
$$\begin{aligned}
\pi^- &\to \mu^- + \overline{\nu}_\mu \\
\pi^+ &\to \mu^+ + \nu_\mu \\
\mu^- &\to e^- + \overline{\nu}_e + \nu_\mu \\
\mu^+ &\to e^+ + \nu_e + \overline{\nu}_\mu
\end{aligned}$$

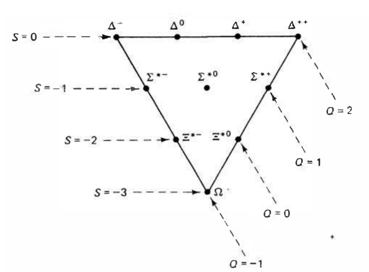
- We also have a *law of conservation of baryon number*; assign to all baryons a baryon number of $A = +1$ and to all the antibaryons $A = -1$.
- Some mesons were discovered; the *Kaon* (K^0), which also has a charged counterpart K^+, and decomposes into $K^0 \to \pi^+ + \pi^-$ and $K^+ \to \pi^+ + \pi^+ + \pi^-$. Then followed the η, ϕ, ω, ρ, and so on.
- Some heavy baryons were discovered as well; the Λ ($\Lambda \to p^+ + \pi^-$), the Σ, Ξ, Δ.
- These new heavy baryons and mesons came to be known as the *strange* particles. They produced on the time scale of $10^{-23}s$, but they decay relatively slowly ($10^{-10}s$). Strange particles are produced by the strong force and decay by the weak force. Strange particles are produced in pairs (so-called *associated production*).
- Gell-Mann and Nishijima assigned to each particle a *strangeness* which is conserved in any strong interaction but not conserved in a weak interaction.
- Later, the a new lepton–the *tau*– and its corresponding neutrino, ν_τ were discovered.

A.2 The Eightfold way

Gell-Mann made periodic tables for baryons and mesons called the *Eightfold Way*. It arranges particles according to their charge and strangeness; we can organise baryons into the *baryon octet* (left). Particles of like electric charge lie along the downward-sloping diagonal lines, and horizontal lines represent particles of like strangeness. Similarly, we have the *meson octet* (middle) and the *baryon decuplet* (right):

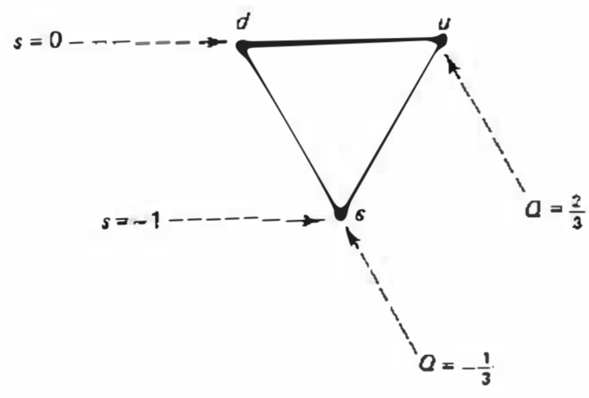


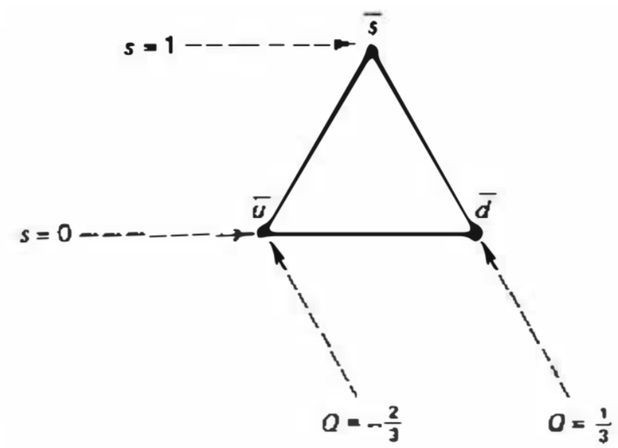




A.3 Quarks

In 1964, Gell-Mann proposed that baryons and mesons are composed of even more elementary constituents, called *qaurks*. Baryons and mesons form *hadrons*: particles made of quarks. Quarks come in many flavors, and every quark has an antiquark of opposite charge and strangeness. Every baryon is composed of three quarks (and every antibaryon if composed of three antiquarks), and every meson is composed of a quark and an antiquark. The three flavors, up, down, and strange, fit into a triangular eightfold way pattern.





The baryon decuplet

qqq	Q	S	Baryon
uuu	2	0	Δ^{++}
uud	1	0	Δ^{+}
udd	0	0	Δ^{0}
ddd	-1	0	Δ^{-}
uus	1	-1	Σ^{*+}
uds	0	-1	Σ^{*0}
dds	-1	-1	Σ^{*-}
uss	0	-2	Ξ^{*0}
dss	-1	-2	Ξ^{*-}
sss	-1	-3	Ω^{-}

The meson nonet

$q\overline{q}$	Q	S	Meson
$u\overline{u}$	0	0	π^0
$u\overline{d}$	1	0	π^+
$d\overline{u}$	-1	0	π^-
$d\overline{d}$	0	0	η
$u\overline{s}$	1	1	K^+
$d\overline{s}$	0	1	K^0
$s\overline{u}$	-1	-1	K^-
$s\overline{d}$	0	-1	$\overline{K}^0$
$s\overline{s}$	0	0	ϕ

Note that the particles in the baryon decuplet above are all spin 3/2. Protons, neutrons, and lambdas, (uud, udd, and uds, respectively) are all spin 1/2, and the Σ^* and Ξ^* particles have corresponding spin 1/2 particles labeled Σ and Ξ respectively, including the corresponding charges.

Quarks also come in three colors (red, green, and blue), and exhibit a Pauli exclusion principle of sorts. A red quark carries one unit of redness, zero blueness, zero greenness; its antiparticle carries minus one unit of redness, and so on. All naturally occuring particles are colorless, meaning that either the total amount of each color is zero or all three colors are present in equal amounts. This rule 'explains' why you can't make a particle out of two or four quarks, and why individual quarks do not appear in nature. Perhaps getting a bit ahead of ourselves, a quark can be seen as a vector in a three-complex-dimensional vector space; color is the basis of this vector space. That is, color is a gauge symmetry; this degree of freedom is designated by the group $SU(3)$.

Later, the c (*charm*), b (*bottom*), t (*top*) quarks were discovered. Particles with these quarks include the Λ_c^+ baryon (udc, spin 1/2), the D mesons ($c\overline{d}$, $d\overline{c}$, $c\overline{u}$, $u\overline{c}$, $c\overline{s}$, $s\overline{c}$ for $D^\pm$, D^0, $\overline{D}^0$, $D_s^\pm$, respectively), and the B mesons ($u\overline{b}$, $b\overline{u}$, $d\overline{b}$, $b\overline{d}$ for $B^\pm$, B^0, $\overline{B}^0$).

A.4 Intermediate vector bosons

The mediator of the weak force is called the *intermediate vector boson*. In the GWS electroweak theory, there are three intermediate vector bosons, two of them charged ($W^\pm$) and one neutral (Z).

A.5 The standard model

- In the current view, all matter is made out of leptons, quarks, and mediators.
- There are six leptons, classified according to their charge, electron number, muon number, and tau number, falling into three generations (first generation for electrons, second for muons, third for tau). Conventionally, we assign a +1 electron/muon/tau number for each electron/muon/tau and the corresponding neutrino, and a −1 for their antiparticles.
- There are six quark flavors, classified by strangeness, charm, beauty, and truth. These quarks also fall into three generations (first is u, d, second is s, c, and third is b, t).

Quark classification

q	Q	D	U	S	C	B	T
d	$-\frac{1}{3}$	-1	0	0	0	0	0
u	$\frac{2}{3}$	0	1	0	0	0	0
s	$-\frac{1}{3}$	0	0	-1	0	0	0
c	$\frac{2}{3}$	0	0	0	1	0	0
b	$-\frac{1}{3}$	0	0	0	0	-1	0
t	$\frac{2}{3}$	0	0	0	0	0	1

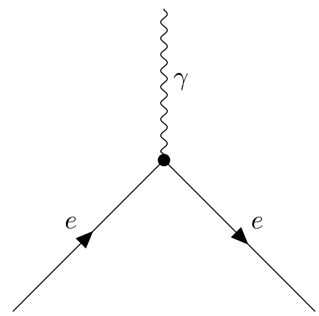
- Every interaction has its mediator–the photon for the electromagnetic force, two W's and a Z for the weak force, the graviton (presumably) for gravity, and the *gluon* for the strong force. In the standard model there are eight gluons, which carry color, and therefore, like quarks, by color confinement, should not exist as isolated particles.
- All ordinary matter is made up of first generation particles. It is believed that there are only three generations compatible with the laws of the universe.
- There's also the Higgs particle, of course.

Appendix B: Elementary forces

The four fundamental forces of nature, in descending strength, are *strong*, *electromagnetic*, *weak*, and *gravitational*. To each belongs a physical theory. Each of these forces is mediated by the exchange of a particle.

B.1 Quantum electrodynamics

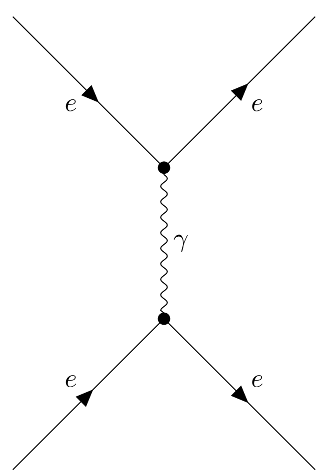
All electromagentic phenomena are ultimately reducible to the following elementary process:



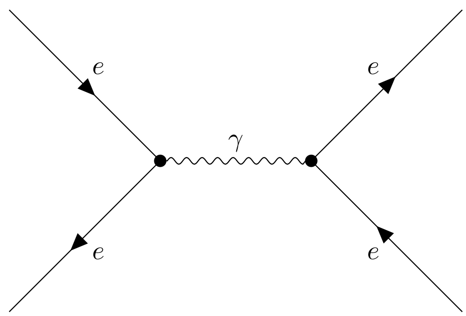
These figures are called *Feynman diagrams*. In these figures, time flows horizontally, to the right, so this diagram reads: a charged particle, e, enters, emits (or absorbs) a photon (γ), and exits. We could say e is an electron, but it could just as well be a quark, or any lepton except a neutrino (the latter is neutral and does not experience an electromagnetic force). You're allowed to twist these Feynman diagrams around into any topological configuration you like. A particle line running backward in time is interpreted as the corresponding antiparticle going forward (the photon is its iwn antiparticle, that's why we don't need an arrow on the photon line). Internal lines which begin and end within the diagram represent particles that cannot be observed without entirely changing the process; these particles are known as *virtual particles*.

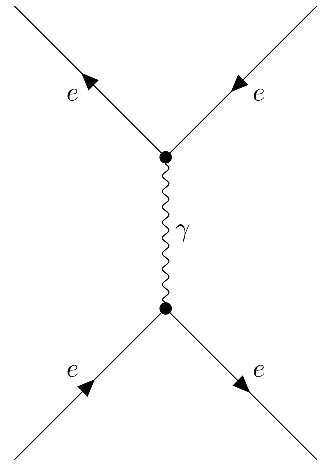
There are a few rules with these tinker-toys; you cannot have a photon line snapped onto an electron line, and you cannot have two photons meet at a vertex.

Example. The process below is *Moller scattering*, and represents what we would call in the classical theory the Coulomb repulsion of like charges.



Example. In this process, known as *Bhabha scattering*, an electron and a positron annihilare ro form a photon, which in turn produces a new electron-positron pair. There are two quite different diagrams representing the same process, both of which must be included in the analysis.

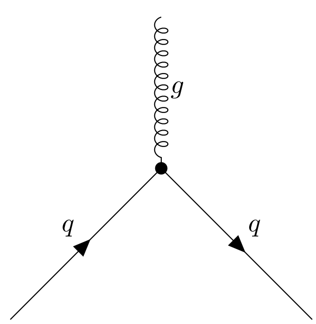




Note. Bhabha and Moller scattering are related by crossing symmetry. In terms of Feynman diagrams, crossing symmetry corresponds to twisting or rotating the figure.

B.2 Quantum chromodynamics

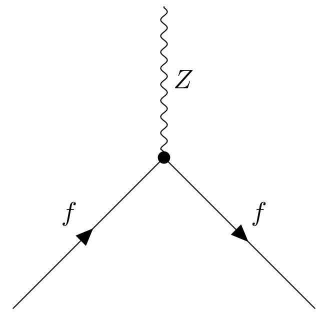
In quantum chromodynamics, color plays the role of charge, and the fundamental process is quark becoming quark plus gluon ($q \to q + g$).



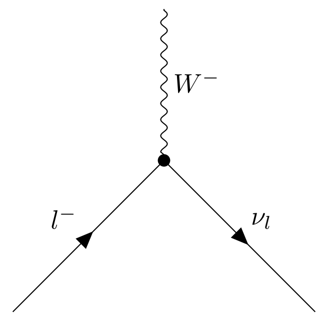
As before, we combine two or more such primitive vertices to represent more complex processes. The differences between chromodynamics and electrodynamics most conspicuously lie in the fact that there are three kinds of color (whereas there is one electric charge). Gluons are 'bicolored', carrying one positive unit of color and one negative unit. Since gluons themselves carry color, they couple directly to other gluons, and hence in addition to the fundamental quark-gluon vertex, we also have primitive gluon-gluon vertices; in fact, two kinds: three-gluon vertices and four-gluon vertices.

B.3 Weak interactions

There are two kinds of weak interactions: charged (mediated by Ws) and neutral (mediated by Z). The fundamental neutral vertex is

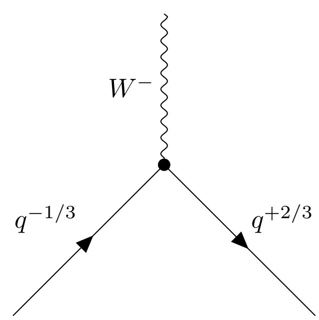


where f is any quark or lepton. The fundamental charged vertex is



Here, a negative lepton l^- converts into the corresponding neutrino, with the emission of a W^- (or absorption of aa W^+: $l^- + \nu_l + W^-$. Notice that leptonic weak vertices connect members of the same generation. In this way, the theory enforces conservation of electron number, muon number, tau number.

While quark interactions mediated by gluons cause color changes, flavor changes are caused by the W bosons. (Since the Z boson is its own antiparticle, possessing no flavor charge, quarks scatter without altering anything except momentum and spin). The fundamental vertex for weak quark interactions is depicted below– the transformation of a quark with charge $-\frac{1}{3}$ into a quark with charge $+\frac{2}{3}$ (by conservation of charge, for a $W^{\pm}$ boson exchange, we need the quark to have a ± 1 change in the charge.



While leptonic interactions are confined to the same generation, we use the *Cabibbo-Kobayashi-Maskawa matrix* (or CKM matrix) to keep track of the weak decays of three generations of quarks. The weak force couples the pairs $\begin{pmatrix} u \\ d' \end{pmatrix}$, $\begin{pmatrix} c \\ s' \end{pmatrix}$, $\begin{pmatrix} t \\ b' \end{pmatrix}$ where d', s', and b' are linear combinations of the physical quarks d, s, and b:

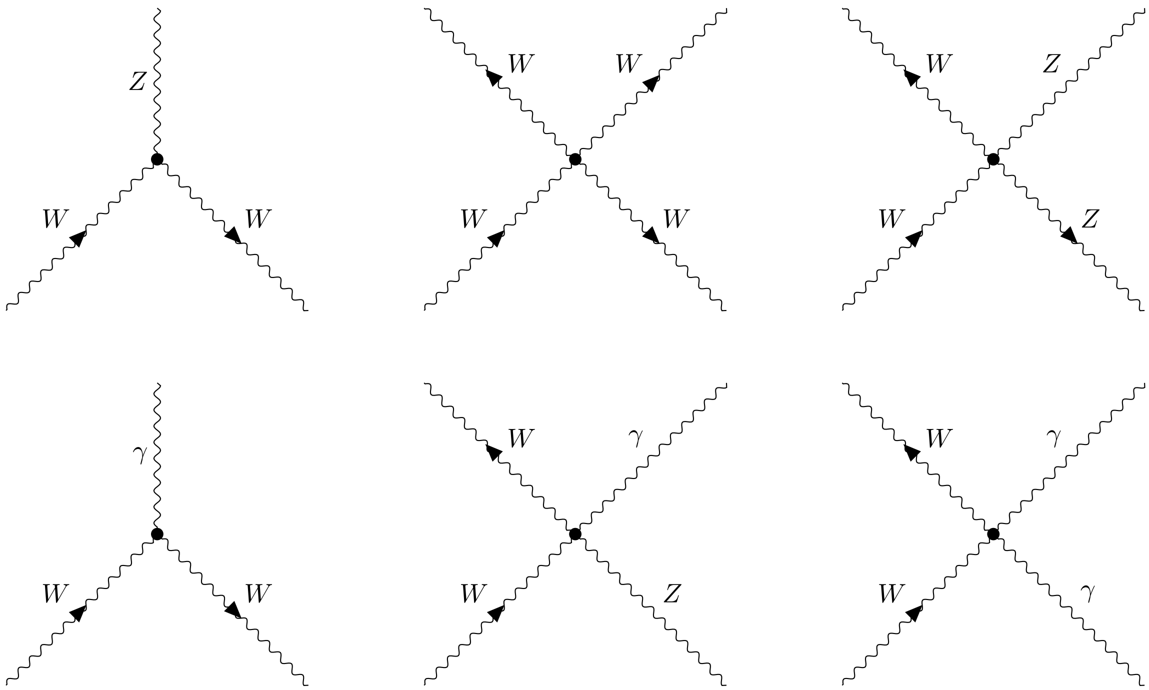
$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

The CKM matrix describes the probabilty of a transition from one flavor j quark to another flavor i quark;

these transitions are proportional to $|V_{ij}|^2$. The individual magnitudes of the CKM elements are approximately

$$\begin{pmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{pmatrix} = \begin{pmatrix} 0.97435 \pm 0.00016 & 0.22500 \pm 0.00067 & 0.00369 \pm 0.00011 \\ 0.22486 \pm 0.00067 & 0.97349 \pm 0.00016 & 0.04182^{+0.00085}_{-0.00074} \\ 0.00857^{+0.00020}_{-0.00018} & 0.04110^{+0.00083}_{-0.00072} & 0.999118^{+0.000031}_{-0.000036} \end{pmatrix}$$

There are also direct couplings of W and Z to another in GWS theory (just as there are direct gluon-gluon couplings in QCD). Moreover, because the W is charged, it couples to the photon. See below the allowed vertices.



B.4 Decays and conservation laws

This summarizes the rules.

(i) *Charge*: All interactions conserve electric charge. In weak interactions, the lepton (or quark) that comes out may not have the same charge as the one that went in, but if so, the difference is carried away by the W.

(ii) *Color*: The electromagnetic and weak interactions do not affect color; at a strong vertex the quark color does change, but the difference is carried off by the gluon. The direct gluon-gluon couplings also conserve color. However, since naturally occuring particles are always colorless, the observable manifestation of color conservation is trivial: zero in, zero out.

(iii) *Baryon number*: In the primitive vertices, if a quark goes in, a quark comes out, so the total number of quarks present is a constant; in this arithmetic we count antiquarks as negative. Of course, we only see baryons (quark number 3), antibaryons (quark number -3), and mesons (quark number 0), so it's more convenient to speak of conservation of baryon number (1 for baryon, -1 for anti). Note that there is no conservation of meson number, since mesons carry zero quark number, so a given collision/decay can produce as many mesons as it likes (consistent with conservation of energy)

(iv) *Lepton number*: In electromagnetic interaction, the same particle comes out (accompanied by a photon) as went in; in a weak interaction if a lepton goes in a lepton comes out (not necessarily the same one)

(v) *Flavor* Flavor is conserved at a strong or electromagnetic vertex, but not at a weak vertex. Because the weak forces are weak, we say that the flavors are approximately conserved.

(vi) *Okubo-Zweig-Iizuka (OZI) rule*: Any strongly occuring process will be supressed if, through only the removal of internal gluon lines, its feynman diagram can be separated into two disconnected diagrams: one containing all of the initial-state particles and one containing all of the final-state particles.

Appendix C: Matrix Lie groups

C.1 Elements

- Define a *matrix Lie group* G to be a subgroup of $GL(n;\mathbb{C})$ (for some n) that, additionally, is closed in the following sense: if $\{g_1, g_2, \cdots\}$ is a sequence of matrices in G converging to an $n \times n$ matrix g, then either g is also in G or g is non-invertible. This last condition is to rule out certain degenerate cases. In other words, we identify the group G with a particular defining, faithful $n \times n$ representation at the outset.

- Obviously, infinite-dimensional Lie groups are not realizable as matrix Lie groups; even some finite-dimensional Lie groups cannot be realized as matrix Lie groups. However, almost all groups of iterest in physics are matrix Lie groups.

We give a few semi-non-trivial examples.

- $O(n,k)$ is the subgroup of $GL(n+k,\mathbb{R})$ that preserves the symmetric bilinear form $\eta = \mathrm{diag}(1, \overset{n}{\cdots} 1, -1, \overset{k}{\cdots}, -1)$ in the sense that $g^T \eta g = \eta$. The group $O(1,3)$ is the group of Lorentz transformations including parity and time-reversal elements.

- $ISO(n,k)$ is the group of isometries of $\mathbb{R}^{n+k}$ equipped with flat metric of signature (n,k); concretely, $ISO(n,k)$ can be parametrized by pairs (Λ, a), where $\Lambda \in O(n,k)$ and $a \in \mathbb{R}^{n+k}$. Namely, $x^{\mu'} = a^\mu + \Lambda^\mu_\nu x^\nu$ for $x^\mu \in \mathbb{R}^{n+k}$. This is evidently a group with group product $(\Lambda, a)\cdot(\Lambda', a') = (\Lambda\Lambda', a + \Lambda a')$.

- $E(n) \equiv ISO(n,0)$ is the n-dimensional *Euclidean group* and $ISO(1,3)$ is called the *Poincare group*.

Theorem. If G is a matrix Lie group and $G_e \leq G$ is the connected component of the identity element e in G, then G_e is a normal subgroup of G.

Some more examples, this time containing some notation that might come later:

- Define $\mathbb{R}^\times \cong GL(1;\mathbb{R})$ the group of nonzero reals equipped with multiplication, and $\mathbb{R}^+ \cong GL^+(1;\mathbb{R})$ the group of all real numbers equipped with addition.

- Define $\mathbb{C}^\times \cong GL(1;\mathbb{C}) \cong U(1) \times GL^+(1;\mathbb{R})$, and define $\mathbb{C}^+ \cong \mathbb{R}^2$.

- Define the *symplectic group* $Sp(n\mathbb{R}) \subset GL(2n;\mathbb{R})$ to be the subgroup of matrices that preserve the *canonical symplectic form* Ω on $\mathbb{R}^{2n}$:
$$\Omega = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$$
That is, $g^T \cdot \Omega \cdot g = \Omega$ for all $g \in Sp(n,\mathbb{R})$. We similarly define $Sp(n;\mathbb{C}) \subset GL(2n;\mathbb{C})$ by $Sp(n;\mathbb{C}) = \{g : g^T \cdot \Omega \cdot g = \Omega\}$. We can then define the *compact symplectic group* $Sp(n) \equiv Sp(n;\mathbb{C}) \cap U(2n)$.

C.2 Matrix exponentials and logarithms

We define the following:

- $\exp(z) = \sum_{k=0}^\infty \frac{z^k}{k!}$, analytic for all $z \in \mathbb{C}$

- $\log(z) = \sum_{k=1}^\infty \frac{(-1)^{k+1}(z-1)^k}{k}$, analytic for $|z-1| < 1$

- $\exp(X) = \sum_{k=0}^\infty \frac{X^k}{k!}$ is the *matrix exponential*, which is an invertible matrix for all $n \times n$ matrices X

- $\log(X) = \sum_{k=1}^\infty \frac{(-1)^{k+1}(X-I)^k}{k}$ is a (not necessarily invertible) $n \times n$ matrix for all X with $||X - 1|| < 1$, where $||\cdot||$ is the *Hilbert-Schmidt norm* defined $||X|| = \left[\sum_{j,k=1}^n |X_{jk}|^2\right]^{1/2}$

- So long as we restrict to the domain where log is defined, $\exp(\log X) = X$ (for $||X - I|| < 1$) and $\log(\exp X) = X$ (for X such that $||\exp X - I|| < 1$).
- The derivative of a matrix exponential is $\frac{d}{dt}\left[e^{Z(t)}\right] = e^{Z(t)}\left\{\left(\frac{I-\exp(-\operatorname{ad}_{Z(t)})}{\operatorname{ad}_{Z(t)}}\right)\left(\frac{dZ}{dt}\right)\right\}$ where we understand the term $\frac{I-\exp(-\operatorname{ad}_{Z(t)})}{\operatorname{ad}_{Z(t)}}$ as an operator expressed as a power series in the operator ad_Z, namely the series obtained by expanding the holomorphic function $\frac{1-e^{-w}}{w}$ of complex variable w around $w = 0$.
- The *Baker-Campbell-Haussdorf (BCH) formula* is

$$\log(e^X e^Y) = X + \int_0^1 dtg\left(e^{\operatorname{ad}_X} e^{t\operatorname{ad}_Y}\right)(Y)$$

for $X, Y \in \mathfrak{g}$, where $g(w)$ is the function

$$g(w) = \frac{\log w}{1 - 1/w} = 1 + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k(k+1)}(w-1)^k$$

which is holomorphic on the disk $|w - 1| < 1$.

Theorem. For any $g \in GL(n;\mathbb{C})$ there exists an $n \times n$ matrix X such that $g = e^X$.

C.3 Matrix Lie algebras

For any $n \times n$ matrix X, define the *1-parameter homomorphism* $\varphi_X : \mathbb{R} \to GL(n;\mathbb{C})$ to be the path $\varphi_X(t) = e^{tX}$ (from the additive group $\mathbb{R}^{(+)}$ into the group of invertible matrices). The set of all points along the route of φ_X is the 1-parameter subgroup of $GL(n;\mathbb{C})$ corresponding to X. Given a matrix Lie group G, we have two equivalent definitions of its *Lie algebra* $\mathfrak{g}$:

- Intuitively, we can define it as the tangent space at the identity element: the set of matrices X such that $X = \frac{d}{dt}\gamma(t)\big|_{t=0}$ for $\gamma : [-1, 1] \to G$ some path with $\gamma(0) = I$.
- More conveniently, $\mathfrak{g}$ is the collection of $n \times n$ matrices X such that $e^{tX} \in G$ for all real t.

$\mathfrak{g}$ is closed under addition, conjugation by elements $g \in G$, multiplication by some $s \in \mathbb{R}$, and the commutator (i.e. $X \cdot Y - Y \cdot X \in \mathfrak{g}$ for all $X, Y \in \mathfrak{g}$). Therefore, the Lie algebra has the structure of a real vector space endowed with a skew-symmetric product. We say that G is a *complex matrix Lie group* when, if $X \in \mathfrak{g}$, then, also, $iX \in \mathfrak{g}$ (where iX means the matrix obtained by multiplying all entries of X by i).

We have to tread carefully in navigating the distinction between *matrix* Lie algebras and *abstract* Lie algebras in the complexificaiton. The *complexification* of a real Lie algebra $\mathfrak{g}$ is accomplished by doubling the underlying vector space V into $V_C \cong V \oplus V = \{(v, w)\}$ with the natural operation of "multiplying by i" given by $J(v, w) = (-w, v)$. The Lie bracket on $V_\mathbb{C}$ is defined as

$$[(v_1, w_1), (v_2, w_2)] = ([v_1, v_2] - [w_1, w_2], [v_1, w_2] + [w_1, v_2])$$

Complexifying a matrix Lie algebra using this construction yields a complex Lie algebra, but we lose the matrix structure. To reconcile this:

Theorem. Suppose $\mathfrak{g}$ is a matrix Lie algebra with the property that, for all $X \in \mathfrak{g}$, the matrix $iX \notin \mathfrak{g}$; then, the (abstract) compactification $\mathfrak{g}_C$ as defined above is isomorphic, as a complex Lie algebra, to the set of matrices that can be expressed in the form $X + iY$ for some $X, Y \in \mathfrak{g}$.

As defined here, the Lie algebra $\mathfrak{g}$ of a matrix Lie group G is a real vector space; this vector space serves as a representation space for *group representation* $\operatorname{Ad} : G \to GL(\mathfrak{g})$ via $\operatorname{Ad}_X Y = [X, Y] \in \mathfrak{g}$ and *algebra representation* $\operatorname{ad} : \mathfrak{g} \to End(\mathfrak{g})$ via $\operatorname{Ad}_g X = g \cdot X \cdot g^{-1} \in \mathfrak{g}$ (for all $X, Y \in \mathfrak{g}$ and $g \in G$). Note that $\operatorname{ad}_X(Y) = \frac{d}{dt}\left[\operatorname{Ad}_{\exp(tX)}\right]_{t=0}$; concretely, $\operatorname{Ad}_{\exp(tX)} Y$ is a matrix that could be expanded in powers of a small, real parameter t, and taking the derivative at $t = 0$ picks out the term linear in t, so we say that Ad "extends" the algebra representation ad to the entire group in the sense that $\operatorname{Ad}_{\exp(X)}(Y) = \left[e^{\operatorname{ad}_X}\right](Y)$. This is a special case of the following general fact:

Theorem. Given a Lie group homomorphism $\Phi : G \to H$, there exits a unique Lie algebra homomorphism $\phi : \mathfrak{g} \to \mathfrak{h}$ between the respective Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ of the Lie groups G and H such that Φ extends to ϕ in the sense that $\Phi\left(e^X\right) = e^{\phi(X)}$ for all $X \in \mathfrak{g}$.

Moreover:

- A ϕ always uniquely induces a local homomorphism f mapping a neighborhood U of the identity in G into a region near the identity of H.
- ϕ may or may not extend to a global homomorphism Φ mapping the entirety of G into H; if such a global Φ does exist, it is unique for connected Lie group.
- A condition that guarantees existence of global Φ is simple-connectedness of G.
- There exists a neighborhood U of $0 \in \mathfrak{g}$ such that the exponential map $\exp : \mathfrak{g} \to G$ (matrix exponential restricted to $\mathfrak{g}$) maps U homeomorphically onto a neighborhood V of the identity I in G.
- In particular, every $g \in V$ can be expressed as $\mathfrak{g} = e^X$ for some $X \in \mathfrak{g}$. Choosing a basis for X allows us to paint coordinates on V. However, for elements of G that are too far from the identity, the series expression for log is ill-defined, so we cannot necessarily do this.
- Let G be a connected Lie group. We have a (not necessarily unique) decomposition of every $g \in G$ as $g = e^{X_1} \cdots e^{X_n}$ for a finite number of matrices $X_1, \cdots, X_n \in \mathfrak{g}$.
- If G is a connected Lie group, the homomorphism $\Phi : G \to H$ inducing a given Lie algebra homomorphism $\phi : \mathfrak{g} \to \mathfrak{h}$ is unique; i.e. if two Lie group homomorphisms induce the same Lie algebra homomorphism, then they are the same Lie group homomorphism.
- We can interpret the BCH formula as endowing a portion of $\mathfrak{g}$ with a group multiplication operation constructed from the Lie bracket, $X * Y = X + Y + \frac{1}{2}[X, Y] + \cdots$ for those $X, Y \in \mathfrak{g}$ such that e^X, e^Y, e^{X*Y} are in the neighborhood U.

This last fact is very deep for reasons that may or may not appear somewhere else in these notes.

C.4 What is needed for physics

Appendix D: Projective representations of Lie algebras and central extensions

- Abstractly, a *projective representation* of a group G on a vector space V over a field F is a group homomorphism from G to the projective linear group $PGL(V) = GL(V)/F^*$, where F^* is the normal subgroup consisting of nonzero scalar multiples of the identity transformation.
- A postulate of quantum mechanics is that state vectors in a separable Hilbert space that are scalar nonzero multiples of each other represent the same pure state; by multiplying state vectors with a phase factor, one obtains a set of vectors called the *ray*. In Weinberg's convention, we represent a system's state by rays $\mathscr{R}_1, \mathscr{R}_2, \cdots$
- *Wigner's theorem*: For any symmetry $\mathscr{R} \to \mathscr{R}'$ of rays, we may define an operator U on the Hilbert space such that if $\Psi \in \mathscr{R}$, then $U\Psi \in \mathscr{R}'$, with U either unitary ($\langle U\Phi, U\Psi\rangle = \langle \Phi, \Psi\rangle$) and linear or else antiunitary ($\langle U\Phi, U\Psi\rangle = \langle \Phi, \Psi\rangle^* = \langle \Phi, \Psi\rangle$) and antilinear ($U(\xi\Phi + \eta\Psi) = \xi^* U\Phi + \eta^* U\Psi$).
- A group of symmetries may be represented projectively on physical states, i.e. elements (say, $T, \overline{T}$) of the symmetry group may be represented on the physical Hilbert space by unitary operators $U(T), U(\overline{T})$ which satisfy the composition rule $U(T)U(\overline{T}) = \exp(i\phi(T,\overline{T})) = U(T\overline{T})$ for some real phase ϕ. We call this a *projective representation*, or a representation "up to a phase".
- Associativity forces $\phi(T_2, T_1) + \phi(T_3, T_2T_1) = \phi(T_3, T_2) + \phi(T_3T_2, T_1)$.
- Any set of functions $\phi(T, \overline{T})$ satisfying the associativity condition and that differ only by functions $\Delta\phi(T, \overline{T}) = \alpha(T\overline{T}) - \alpha(T) - \alpha(\overline{T})$ is called *two-cocycle*.
- A projective representation with phases of the form of $\Delta\phi(T, \overline{T})$ can be replaced with an ordinary representation by replacing $U(T)$ with $\tilde{U}(T) = U(T)\exp(i\alpha(T))$ for which $\tilde{U}(T)\tilde{U}(\overline{T}) = \tilde{U}(T\overline{T})$.
- We are interested in whether a symmetry group allows any nontrivial two-cocycles; that is, whether it may have a representation on the physical Hilbert space that is intrinsically projective, in the sense that the phase $\phi(T, \overline{T})$ cannot be eliminated.
- Consider the effect of a phase on the commutation relations of the generators of infinitsimal transformations. When either $\overline{T}$ or T is the identity, the phase vanishes. Using coordinates θ^a to parametrize group elements, the expansion of ϕ about $\theta = \overline{\theta} = 0$ must be of the form

$$\phi(T(\theta), T(\overline{\theta})) = f_{ab}\theta^a\overline{\theta}^b$$

  where f_{ab} are real numerical constants; inserting this expansion into the power series of expansion of the composition rule, we are led to $[t_b, t_c] = iC^a_{bc}t_a + iC_{bc}1$, where C_{bc} is the antisymmetric coefficient $C_{bc} = -f_{bc} + f_{cb}$. The appearance of the terms on the right-hand side of the commutation relation proportional to the unit element (so-called *central charges*) is the counterpart for the Lie algebra of the presence of phases in a projective representation of a group. We obtain constraints following from the Jacobi identity:

$$C^a_{bc}C^e_{ad} + C^a_{cd}C^e_{ab} + C^a_{db}C^e_{ac} + = 0$$
$$C^a_{bc}C_{ad} + C^a_{cd}C_{ab} + C^a_{db}C_{ac} + = 0$$

  For solutions $C_{ab} = C^e_{ab}\phi_e$, we can eliminate the central charges from our commutation relation by a redefinition of the generators $t_a \to \tilde{t}_a = t_a + \phi_a$, giving $[\tilde{t}_b, \tilde{t}_c] = iC^a_{bc}\tilde{t}_a$.
- The phase of any representation $U(T)$ of a given group can be chosen so that $\phi = 0$ in the composition relation if two conditions are met:
  (a) The generators of the group in these representation can be redefined so as to eliminate all central charges from the Lie algebra.

(b) The group is simply connected (any two group elements may be connected by a path lying within the group, and any two such paths may be continuously transformed into one another; equivalently, any loop that starts and ends at the same group element may be shrunk continuously to a point).

- Recall that a Lie algebra is *semi-simple* if it has no 'invariant Abelian' subalgebra, consisting of generators that commute with each other and whose commutators with any other generators also belong to that subalgebras.
- For the first condition (a), note that any central charges in semi-simple Lie algebras may always be removed by a redefinition of generators. While the Poincare group is not semi-simple (while the algebra of the angular momentum generators is, the momentum generators form an invariant Abelian subalgebra), we in fact can perform a redefinition of generators to remove central charges, which I will not state because there are too many symbols. Therefore it satisfies the first condition.
- The second condition (b) concerns the topology.

Appendix E: Superanalysis

- Let ζ^a, $a = 1, \cdots, N$ be a set of generators for an algebra, which anticommute:
$$\zeta^a \zeta^b = -\zeta^b \zeta^a, \quad (\zeta^a)^2 = 0$$
for all a, b. The algebra is called a *Grassmann algebra* and denoted by Λ_N. In what follows the formal limit $N \to \infty$ is always taken, and the elements of the algebra, when expressed in terms of the ζ^a, are formal infinite series.

- The elements of Λ_∞ are called *supernumbers*. Every supernumber can be expressed in the form $z = z_B + z_S$, where z_B is an ordinary complex number (called the *body* of z) and
$$z_s = \sum_{n=1}^{\infty} \frac{1}{n!} c_{a_1 \cdots a_n} \zeta^{a_n} \cdots \zeta^{a_1}$$
is the *soul* of z, the c's also being complex numbers. The c's are completely antisymmetric in their indices, and summation over repeated indices is to be understood.

- Every analytic function on the complex numbers may be extended to a supernumber-valued function on Λ_∞ by the formal series
$$f(z) := \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(z_B) z_S^n$$
Let $\pi : \Lambda_\infty \to \mathbb{C}$ be the mapping that replaces each supernumber by its body z_B, and let z_0 be a singular point of $f|_{\pi(\Lambda_\infty)} = f|_{\mathbb{C}}$. Then every element of $\pi^{-1}(z_0)$ is a singular point of f.

- One may consider matrices whose elements are supernumbers. The body of a matrix is the ordinary matrix obtained by taking the body of each of its elements, and the soul of the matrix is its remainder. A square matrix has an inverse iff its body is nonsingular; the inverse is unique and expressible as a formal infinite series
$$M^{-1} = M_B^{-1} - M_B^{-1} M_S M_B^{-1} + M_B^{-1} M_S M_B^{-1} M_S M_B^{-1} - \cdots$$

- Every supernumber may be split into its even and odd parts:
$$\begin{aligned} z &= u + v \\ u &= z_B + \sum_{n=1}^{\infty} \frac{1}{(2n)!} c_{a_1 \cdots a_{2n}} \zeta^{a_{2n}} \cdots \zeta^{a_1} \\ v &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} c_{a_1 \cdots a_{2n+1}} \zeta^{a_{2n+1}} \cdots \zeta^{a_1} \end{aligned}$$

- Superenumbers that are either purely odd or purely even are called *pure*. Odd supernumbers anticommute among themselves and are called *a-numbers*; even supernumbers commute with everything and are called *c-numbers*. The set of all c-numbers is a commutative subalgebra of Λ_∞ denoted by $\mathbb{C}_c$, and the set of all a-numbers is denoted by $\mathbb{C}_a$; its is not a subalgebra. The square of every a-number vanishes.

- To define real supernumbers one extends the rules of complex conjugation from $\mathbb{C}$ to Λ_∞ by adding the relations
$$\begin{aligned} \zeta^{a*} &= \zeta^a \\ (z + z')^* &= z^* + z'^* \\ (zz')^* &= z'^* z^* \end{aligned}$$
for all a and for all $z, z' \in \Lambda_\infty$. z is said to be real if $z^* = z$, imaginary if $z^* = -z$, and complex otherwise.

- The sets of real elements of $\mathbb{C}_c$ and $\mathbb{C}_a$ are denoted by $\mathbb{R}_c$ and $\mathbb{R}_a$ respectively. The product of two real c-numbers is a real c-number; the product of a real c-number and a real a-number is a real a-number; the product of two real a-numbers is an imaginary bodiless c-number.

Appendix F: Clifford algebras

The purpose of this section is to understand spinors from their Clifford algebra structure. This is mostly a summary of *The Spinorial Chessboard*. In what follows, all vector spaces and algebras considered are over the field K of real or complex numbers. The vector spaces are finite-dimensional and algebras are unital associative algebras (i.e. there is a multiplicative identity, and multiplication is associative).

Much of mathematics involves the idea of linearization. The tensor product of vector spaces is introduced in order to represent any bilinear map as a linear map composed with the canonical bilinear map. The derivative of a differentiable map is, in a well-known sense, its linear part. Clifford algebras can be viewed as allowing a linearlization of quadratic form.[16]

F.1 First construction

- Let g denote a scalar product in a vector space V over K and let A be an algebra over the same field. A *Clifford map* of (V, g) into A is a linear map $f : V \to A$ such that, for any $u \in V$, $f(u)^2 = g(u, u)$.

- Let $T(V) = \bigoplus_{i=0}^{\infty} \overset{i}{\otimes} V$ be the tensor algebra of V and let $I(g)$ be its two-sided ideal generated by all elements of the form $u \otimes u - g(u, u)$, where $u \in V$. The *Clifford algebra* $Cl(g)$ of (V, g) is the factor algebra $T(V)/I(g)$.[17]

- The canonical map $\kappa : T(V) \to Cl(g)$ restricted to the subspace V of $T(V)$ is a Clifford map.

- The Clifford map $\kappa : V \to Cl(g)$ is universal, i.e. if $f : V \to A$ is a Clifford map of (V, g) into A, then there exists a unique homomorphism of algebras with unity $\tilde{f} : Cl(g) \to A$ such that $f = \tilde{f} \circ k$.

- $\kappa : V \to Cl(g)$ contains not only information about the algebra, but also on how the vector space is embedded in the algebra and what is the scalar product. It should be understood that we understand Clifford algebras together with their underlying vector space V and the Clifford embedding κ.

- One can write $uv + vu = 2g(u, v)$ for every $u, v \in V$; observe that two vectors anticommute iff they are orthogonal.

- Cl is a functor; for two vector spaces with scalar products $(V_1, g_1), (V_2, g_2)$, if $f : V_1 \to V_2$ is linear and $g_2(f(u), f(v)) = g_1(u, v)$ for every $u, v \in V$, then there exists a homomorphism of algebras with unity $Cl(f) : Cl(g_1) \to Cl(g_2)$ such that $Cl(f) \circ \kappa_1 = \kappa_2 \circ f$, where $\kappa_i : V_i \to Cl(g_i)$ is the restriction to V_i of the canonical map.

- In particular, to any orthogonal transformation $f \in O(g)$ there corresponds an automorphism $Cl(f)$ of the algebra $Cl(g)$ and there is a homomorphism of groups $Cl : O(g) \to Aut(Cl(g)$.

- If f is the reflection, $f(v) = -v$, $v \in V$, then $\alpha = Cl(f)$ is called the *main automorphism* of the Clifford algebra. When it is necessray to specify the vector space V underlying the Clifford algebra where the main automorphism is define, we denote it $\alpha(V)$. It is involutive and defines the basic $\mathbb{Z}_2$-grading of the Clifford algebra, $Cl(g) = Cl_0(g) \oplus Cl_1(g)$, where the decomposition of $a \in Cl(g)$ into its even $a_0 \in Cl_0(g)$ and odd $a_1 \in Cl_1(g)$ parts is given by $a_0 = \frac{1}{2}(a + \alpha(a))$, $a_1 = \frac{1}{2}(a - \alpha(a))$.

- The transposed (opposite) algebra of $Cl(g)$, which we denote by A, has un underlying vector space the same as that of $Cl(g)$, and the product of a and b in A is equal to the product of b and a in $Cl(g)$. The map $f : V \to A$ given by $f(u) = u$) is a Clifford map and $\tilde{f} : Cl(g) \to A$ is an isomorphism of algebras with unity which may be represented as an antiisomorphism $\beta : Cl(g) \to Cl(g)$. It is characterized as a linear isomorphism such that $\beta(1) = 1$, $\beta(u) = u$, and $\beta(ab) = \beta(b)\beta(a)$ for every $u \in V$ and $a, b \in Cl(g)$.

[16] Recall the definition of a quadratic form. Let R be a commutative ring and M a (left) R-module. A map $Q : M \to R$ is a *quadratic form* on M if i) $Q(\alpha x) = \alpha^2 Q(x)$ for $\alpha \in R$, $x \in M$, and ii) the map $\Phi : M \times M \to R$ defined $(x, y) \mapsto Q(x + y) - Q(x) - Q(y)$ is a blinear form. Φ is called the bilinear form associated to Q.

[17] This "Clifford map" construction is not universal; for example, Bourbaki constructs a Clifford algebra in terms of quadratic form Q on an R-module M, defining $Cl(Q) = T(M)/(x \otimes x - Q(x) \cdot 1$ $(x \in M)$.

The map β will be called the *main antiautomorphism* of $Cl(g)$, and we will sometimes indicate the underlying vector space of the algebra by writing $\beta(V)$.

F.2 Vector space structure

- Recall that the *exterior algebra*, or *Grassmann algebra*, of a vector space V is defined to be the quotient algebra of the tensor algebra $T(V)$ by the two-sided ideal generated by all elements of the form $x \otimes x$ such that $x \in V$.

- For every $u \in V$, let j_u be the endomorphism of $\bigwedge V$ given by $j_u(w) = u \wedge w$. If $\omega \in V^*$, then the interior product i_ω is an antiderivation of $\bigwedge V$ characterized by $i_\omega(u) = \langle \omega, u \rangle$ for every $u \in V$, and $i_\omega(v \wedge w) = i_\omega(v) \wedge w + (-1)^\ell v \wedge i_\omega(w)$ for $v \in \bigwedge^\ell V$ and $w \in \bigwedge V$. Since $i_\omega \circ j_u + j_u \circ i_\omega = \langle \omega, u \rangle$ id, the map $u \mapsto f(u) = i_{g(u)} + j_u$ is a Clifford map from V to $End \bigwedge V$, therefore extending to a homomorphism of algebras with unity $\tilde{f} : Cl(g) \to End \bigwedge V$.

- As vector spaces, the Clifford algebra of (V, g) and the Grassman algebra $\bigwedge V$ are isomorphic. Particularly, the map $F_g : Cl(g) \to \bigwedge V$ given by $F_g(s) = \langle \tilde{f}(s), 1 \rangle$ is an isomorphism of vector spaces which preserves the $\mathbb{Z}_2$-grading and is natural with respect to isometries. This last property means that, if $f : V \to W$ is an isometry from (V, g) to (W, h), then $(\bigwedge f) \circ F_g = F_h \circ Cl(f)$, where $\bigwedge f : \bigwedge V \to \bigwedge W$ is the extension of f, $f(1) = 1$, $(\bigwedge f)(v_1 \wedge \cdots \wedge v_\ell) = f(v_1) \wedge \cdots \wedge f(v_\ell)$.

- Let (e_i), $i = 1, \cdots, n$ be an orthogonal basis in (V, g). Then F_g maps $e_{i_1} e_{i_2} \cdots e_{i_\ell} \in Cl(g)$, with $1 \leq i_1 < i_2 < \cdots < i_\ell \leq n$ and $1 \leq l \leq n$, into $e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_\ell} \in \bigwedge V$. Therefore, the set $e_{i_1} e_{i_2} \cdots e_{i_\ell}$ supplemented with 1 (the product of an empty sequence of basis vectors) constitutes a linear basis of $Cl(g)$ consisting of 2^n elements; the set $\{e_1, \cdots, e_n\}$ generates $Cl(g)$ as an algebra.

- F_g allows us to refer to the elements of the Clifford algebra by the names of their images in $\bigwedge V$. Thus $uv - vu$, where $u, v \in V$, is a bivector and $\eta = e_1 e_2 \cdots 2_n$ is a volume element.

- The main antiautomorphism β, being linear and involutive, defines a decomposition $Cl(g) = Cl_+(g) \oplus Cl_-(g)$, corresponding to the decomposition $a = a_+ + a_-$ of $a \in Cl(g)$, where $a_\pm = \frac{1}{2}(a \pm \beta(a))$ so that $\beta(a_\pm) = \pm a_\pm$. The dimension of $Cl_+(g)$ depends only on the dimension n of V; $\dim Cl_+(g) - \dim Cl_-(g) = 2^{(n-2)/2+1}\left(\cos\frac{n\pi}{4} + \sin\frac{n\pi}{4}\right)$.

F.3 Complements on $\mathbb{Z}_2$-graded algebras

- An algebra A over K is said to be $\mathbb{Z}_2$-*graded* or simply *graded* if there is a decomposition of A into a direct sum of vector spaces $A = A_0 \oplus A_1$ such that $A_p A_q \subseteq A_{p+q}$ where the sum $p + q$ is being taken mod 2.

- The elements of A_0 (resp. A_1) are said to be even (resp. odd). Sometimes one writes $\deg a = p$ if $a \in A_p$.

- To describe a graded algebra, it is often sufficient to indicate the embedding $A_0 \to A$.

- A *graded derivation* of degree $p \in \mathbb{Z}_2$ is a linear map $d : A \to A$ such that $dA_q \subset A_{p+q}$ and $d(ab) = (da)b + (-1)^{pq} adb$ for every $a \in A_q$ and $b \in A$. A graded derivation of degree 0 is simply called a *derivation*, while a graded derivation of degree 1 is an *antiderivation*.

- A *graded Lie algebra* (or *super Lie algbera*) is a graded algebra $\mathfrak{L} = \mathfrak{L}_0 \oplus \mathfrak{L}_1$ with a product $[,]$ which is "superanticommutative",
$$[a, b] = (-1)^{pq+1}[b, a]$$
for every $a \in A_p$, $b \in A_q$, and such that the map $\mathrm{ad}(a) : \mathfrak{L} \to \mathfrak{L}$, where $\langle \mathrm{ad}(a), b \rangle = [a, b]$ is a graded derivation of degree $p = \deg a$.

- With any graded algebra A one associates the graded Lie algebra $\mathrm{Der}\, A$ of derivations by defining

$$d_p, d_q] = d_p \circ d_q(-1)^{pq+1} d_q \circ d_p$$

where d_p, d_q are graded derivations of degree p and q, respectively.

- If the graded algebra A is associative, then one can form the graded Lie algbera $\mathfrak{L}(A)$; its underlying vector space and grading are those of A and the product is given by the graded (super)commutator,

$$[a, b] = ab + (-1)^{pq+1} ba$$

where $a \in A_p, b \in A_q$.

- The *supercenter* $SZ(A)$ of a graded associative algebra A is the set

$$\{a \in A | [a, b] = 0 \ \forall \ b \in A\}$$

with the bracket as defined above.

- If $SZ(A) = A$ then A is said to be *supercommutative.*

- If A has unity 1, then $1 \in A_0$; in this case, A is said to be *supercentral* if its supercenter coincides with the number field $K \subset A_0$.

Example. The algebra $\mathbb{C}$ is a graded associative algebra over $\mathbb{R}$; the grading is defined by declaring pure imaginary numbers to be odd. This algebra is supercentral, but not supercommutative (e.g. $[i, i] = -2$). It is not central over $\mathbb{R}$.

There are a number of standard constructions which lead to graded algebras; in what follows, take A and B to denote a graded and ungraded algebra, respectively.

(i) The *double* of B is the graded algebra $B \to 2B$ where $2B = B \oplus B$ and the elements of the form $(b, -b), b \in B$, are odd.

(ii) The tensor product $A \otimes B$ has the standard grading $(A \otimes B)_p = A_p \otimes B$, where $p = 0, 1$.

(iii) If both A, A' are graded, then their *graded tensor product* $A \otimes A'$ has $A \otimes A'$ as its underlying vector space,

$$\deg(a \otimes a') = \deg a + \deg a' \bmod 2$$

and

$$(a \otimes a'_p) \cdot (b_q \otimes b') = (-1)^{pq} ab_q \otimes a'_p b'$$

for every $a \in A$, $b_q \in A_q$, $a'_p \in A'_p$, and $b' \in A'$.

(iv) If A is graded, then the opposite algebra has the same underlying vector space structure and grading as A, and an opposite product $\times$ defined by $a \times b = (-1)^{pq} ba$ for $a \in A_p$ and $b \in A_q$. Therefore, A is supercommutative iff the identity map from A to its opposite is an isomorphism of graded algebras.

F.4 Graded structure

- If (e_i) is an orthogonal basis in V, then the volume elemnt satisfies $\eta u + (-1)^n u \eta = 0$ for every $u \in V$. For n odd, $K \oplus K\eta$ is contained in the center of $Cl(g)$. More precisely, let $Z_\pm(g) = \{z \in Cl(g) | zu \mp uz = 0 : \ \forall u \in V\}$. If n is even, then $Z_+(g) = K)$, $Z_-(g) = K\eta$. If n is odd, then $Z_+(g) = K \oplus K\eta$, $Z_-(g) = \{0\}$.

- Every Clifford algebra is supercentral.

- The Clifford algebra of an even-dimensional space is central.

- The center of a Clifford algbera of an odd-dimensional space is generated by the volume element.

Theorem. (*Chevalley*) Consider the direct sum $(V \oplus W, g \oplus h)$ of two vector spaces with scalar products, $(V, g), (W, h)$. The map $f : V \oplus W \to Cl(g)\widehat{\otimes}Cl(h)$ given by $f(v, w) = v \otimes 1 + 1 \otimes w$ is linear and has the Clifford property $f(v, w)^2 = g(v, v) + w, w)$. There is a natural, grading-preserving isomorphism $\tilde{f} : Cl(g \oplus h) \to Cl(g)\widehat{\otimes}Cl(h)$of algebras with unity resulting from extending f.

- If g is a scalar product in V, then $-g$ is the opposite scalar product in V, $(-g)(u, v) = -g(u, v)$. The injection $V \to Cl(g)^{opp}$ is a Clifford map $(V, -g) \to Cl(g)^{opp}$. Therefore:

- The algebras $Cl(-g)$ and $Cl(g)^{opp}$ are isomorphic as graded algebras with unity.

Chapter 2

Mirror Symmetry

1 Introduction for the mathematician

- The starting point for defining a quantum field theory is the choice of a manifold M of dimension d. For most, but not all, QFTs, the manifold is viewed as a Riemannian manifold with a smooth metric on it. For many physical applications, we consider manfiolds with $d-1$ positive directions and one negative directions of the metric, as in the d-dimensional Minkowski space. M may or may not have boundaries; in the case it does, some additional information is needed at the boundaries to define the QFT.
- In the QFT, one aims to integrate over the space parametrizing some fields. The operation of integration over the fields is the *path integral*. For example, we may consider a principal bundle over M with a connection (the choice of connection is called *picking a gauge field*). We may also be considering sections of a vector bundle over M (these fields are called *matter fields*).
- We may also consider the space of maps $X : M \to N$ for some target manifold N; the field theories associated with integrating over the space of such maps are called *sigma models*.
- Sometimes we may be interested in considering various choices of metrics on M; integrating over such choices is *quantum gravity*.
- In integrating over the field space, we have to choose a measure on it; there is often a natural choice. The measure is usually weighted by $\exp(-S)$, where S is the action functional on the space of fields in question; in the Minkowski signature the measure is modified by the weight $\exp(iS)$.
- One can consider the case where M has some boundary components $\partial M = \cup_i B_i$, which can only occure when $\dim M \geq 1$. In such case, in defining the integration over the field space we have to specify boundary conditions for fields on B_i. The space of field configurations on each B_i gives rise to a Hilbert space $\mathcal{H}_i$, and the path integral can be viewed as a multilinear map $\otimes_i \mathcal{H}_i \to \mathbb{C}$. If we glue two manifolds along their boundaries, the path integral can be performed by pairing the states corresponding to the boundaries that were glued.
- In the case $M = N \times I$, where N is a manifold without boundaries and I is an interval of length T, the path integral gives rise to a linear map $U(T) : \mathcal{H} \to \mathcal{T}$ by dualizing the Hilbert space corresponding to one of the boundaries. We learn that $U(T_1)U(T_2) = U(T_1 + T_2)$, in turn defining an operator H as the generator of U, $U(T) = \exp(-TH)$ in the Euclidean case, or $U(T) = \exp(-iTH)$ in the case where I corresponds to the negative direction in the signature (the "time"). H is called the *Hamiltonian* and is (usually) a Hamiltonian operator.

2 QFT in $d = 0$

- For the remainder of the section, take M to be a point.

- The simplest case of a QFT is taking the field X to correspond to maps $M \to \mathbb{R}$; the *partition function* is an integral
$$Z = \int dX e^{-S[X]}$$
The *correlation functions* are just weighted integrals
$$\langle f(x) \rangle = \int dX f(X) e^{-S[X]}$$
- We can also determine correlation functions by deforming the action $S \mapsto S' = S + \sum_i a_i f_i(X)$. Then $\langle f_i(X) \rangle = \frac{\partial Z(\alpha, a_i)}{\partial a_i}\Big|_{a_i=0}$ where α is a parameter of S and $Z(\alpha, a_i = \int dX e^{-S'(X)}$.
- For example, for a toy model with $S[X] = \frac{\alpha}{2} X^2 + i\epsilon X^3$, let us write $Z(\alpha, \epsilon)$. Trivially, $Z(\alpha, 0) = \sqrt{\frac{2\pi}{\alpha}}$. We often define the normalization of the measure of the measure of integration such that we get rid of the factor $\sqrt{\frac{2\pi}{\alpha}}$, i.e. we consider the normalized partition function $\frac{Z(\alpha,\epsilon)}{Z(\alpha,0)}$.
- If $\epsilon << 1$, we can expand the partition function in powers of ϵ to obtain a *perturbative expansion*
$$Z = \int dX e^{-\frac{\alpha}{2}X^2 - i\epsilon X^3} = \int dX \sum_{n=0}^{\infty} e^{-\frac{\alpha}{2}X^2} \frac{(-i\epsilon X^3)^n}{n!}$$
We assume that the perturbative expansion exists and do not worry about issues of convergence.
- *Feynman diagrams* are useful for perturbative computations in QFTs.[1] Consider the function $f(\alpha, J) = \int e^{-\frac{\alpha}{2}X^2 + JX}$; J is known as the source in physics. The integral is trivially $f(\alpha, J) = \sqrt{\frac{2\pi}{\alpha}} e^{\frac{J^2}{2\alpha}}$. Using $f(\alpha, J)$ we can write down some other useful integrals as derivatives of this function, in particular $\int X^r e^{-\frac{\alpha}{2}X^2} dX = \frac{\partial^r f(\alpha, J}{\partial J^r}\Big|_{J=0}$. Pairs of $\frac{\partial}{\partial J}$ act together for a non-vanishing contribution to the above quantity, which can be seen from the form $f \propto \exp(J^2/2\alpha)$; first, $\frac{\partial}{\partial J}$ brings down a term $\frac{J}{\alpha}$ from the exponent, then another $\frac{\partial}{\partial J}$ absorbs it. Since each $\frac{\partial}{\partial J}$ corresponds to an X, we see in computing the integral of X^r with the Gaussian measure that we have to consider all ways of choosing pairs of them. This operatation when used for computing such integrals is called choosing pairs and contracting them; this contraction is also called *Wick contraction*. Each pair gives a factor of $\frac{1}{\alpha}$ and therefore $\frac{\partial^r f(\alpha, J)}{\partial J^r}$ gives $\left(\frac{1}{\alpha}\right)^{r/2} \times$ (#of ways of contracting).
- Sometimes we draw lines to show possible contractions; such a line is called a *propagator*, and therefore each propagator is weighted with a factor of $\frac{1}{\alpha}$.
- For the first nontrivial correction to $Z(\alpha, 0)$, $\mathcal{O}(\epsilon)^2 : \frac{(-i\epsilon)^2}{2!} \int dX\ X^3 \times X^3 \times e^{-\frac{\alpha}{2}X^2}$. In general, a term of the form X^k with $k > 1$ leads to a vertex with k edges emanating from it. So for our $\mathcal{O}(\epsilon)^2$, we find the way to contract pairs of a vertex with three edges emanating by contracting edges; we find that there are two possible contracted graphs, and 15 possible pairs of contractions. This can easily be shown on paper. These contracted graphs are the Feynman diagrams. In general, we obtain both connected and disconnected diagrams.
- $Z(\alpha, \epsilon) = e^{\sum \text{connected graphs}}$. Moreover, the combinatorial factor associated to each connected graph G is given by $\frac{(-3! i\epsilon)^V \alpha^{-E}}{|Aut(G)|}$ where V is the number of vertices, E is the number of edges, and $|Aut(G)|$ denotes the order of the automorphism group of the graph.
- $F = \ln Z$ is usually called the free energy and is usually given by minus the sum of the connected graphs.

[1] It is not necessary in this rather simple example, but it setting it up helps in undersanding Feynman diagrams in higher-dimensional QFTs.

Chapter 3

Bibliography

The following texts were used throughout these notes.

- *Introduction to Elementary Particles* by Griffiths
- *A Standard Model Workbook* by Moore
- *An Introduction to Quantum Field Theory* by Peskin & Schroeder
- *Quantum Field Theory & the Standard Model* by Schwartz
- *Lie Algebras, Lie Groups, and Representations* by Hall
- *The Quantum Theory of Fields* by Weinberg
- *Local Quantum Physics* by Haag
- *The Spinorial Chessboard* by Budinich & Trautman
- *Elements de Mathematique: Algebre* by Bourbaki
- *The Global Approach to Quantum Field Theory* by deWitt
- *Electromagnetic Duality for Children* by Figueroa-O'Farrill
- *Introduction to Quantum Mechanics* by Griffiths[1]

[1]We don't use this one much, so when referring to "Griffiths", we're generally talking about the particles book.